



Linnæus University

Sweden

Bachelor Thesis

“Do you see it now?”

A qualitative study of how visually impaired's retail experience gets affected in supermarkets.



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Abstract

Since retailing conceptualisation, it has been at the forefront of societal distribution of necessities, goods and experiences. Human beings, the consumers of retail, have daily or multiple times used supermarkets to express their purchase behaviour and elevate their consumer lives. In the retail space itself the consumer uses their senses of visual, auditory, olfactory and touch to accurately perceive all around them. But there is a group of individuals who face various challenges in the retail experience: people who are visually impaired. For them, the retail experience differs significantly from the norm. Thus, the purpose of this study is to explore how people who are visually impaired experience the retail setting of supermarkets. With an aim to analyse how in-store design and sensory cues can be used to enable and affect supermarkets for the visually impaired. Additionally, the impact of using aids will be explored to facilitate their role in retail experience.

Empirical findings were collected with the help of qualitative single case studies with semi structured interviews. In total, one pilot-study was conducted and 15 visually impaired people were interviewed in Sweden. The informants were asked about their retail experience in supermarkets and how in-store design, senses and aids affect their personal experiences. The findings were analysed and coded through thematic analysis and later discussed with various peer-reviewed literature.

Conclusively, this thesis found that visually impaired people experience a significant difference from sighted people in supermarkets. Facing numerous challenges and obstacles in their visits. It was found that visually impaired people are separated from other customer segments through various inaccessible factors in the retail setting. However, as these factors work as barriers, the research shows that there exist solutions and potential in the supermarket setting, extending through in-store design, sensory cues, and aids. By facilitating and developing the areas, major improvements to customer accessibility can be introduced.

Key words

Visually impaired, Retail experience, In-store design, Senses, Aids for visually impaired, Supermarkets, Accessibility



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1. Introduction

In this chapter, an introduction of the main topics which are covered in this thesis is presented by the authors. First, the background provides the needed knowledge of the topics. Secondly, a problem discussion of these topics' relevance from both a theoretical and practical perspective, resulting in the identification of the research gap. Lastly, the research questions, the purpose as well as the delimitations of this thesis is presented.

1.1 Supermarkets & visually impaired

1.1.1 Retailing & Supermarkets

Retailing, one of humanity's oldest business practices, has transformed from rudimentary food and weapon exchanges in prehistoric times (Tiwari, 2009) to today's emphasis on unique customer experiences (Levy, Weitz & Grewal, 2018), navigating a dynamic retail landscape of evolving trends (Kohan, 2022). At its core, retailing involves businesses enhancing the value of products or services through various operational activities within the context of consumer purchases for personal use (Levy, Weitz & Grewal, 2018). Central to this process is the supply chain, including all parts from product creation to customer delivery. Retailers, serving as intermediaries between manufacturers and consumers, play a crucial role in this chain. Multisensory management in retail environments presents a formidable challenge (Mau, Schweizer, & Oriet, 2023), but technological advancements offer opportunities for innovative store design, catering to diverse customer preferences and intensifying the retailer's responsibility for crafting memorable and well-developed retail experiences. Retailers can influence consumers' perceptions of products by modifying the multisensory aspects of the store environment (Imschloss & Kuehn, 2017).

The retail experience is a theoretical concept deeply intertwined with the consumer as well as their impression and perception. For example, regarding the buying process of a product, there are two perspectives in play: the retailer and the experience they offer the customer, and the journey customers go through while purchasing (Invergehq, n.d.). The retail experience is the result of customer interactions with the business, including pre-, during and after the purchase. When creating the experience, it is evident that the customer does not necessarily buy only the product or the service but the experience itself (Srinivasan & Srivastava, 2010). Therefore, having a broader understanding of the retail experience and the customer journey is the key to understanding the relationship which can manifest (Poorrezaei, Pich & Resnick, 2023).

Among the myriad forms of retail establishments, supermarkets represent a significant sector. Levy, Weitz, and Grewal (2018) outline the primary



characteristics of a traditional supermarket as a spacious, self-service retail food store offering both consumables along with non-consumable products. Given that consumers typically make at least three weekly trips for consumables compared to only one trip for non-consumable items. Supermarkets prioritise being food retailers first to be able to offer commerce items like fresh dairy products, flowers, and meat.

1.1.2 Consideration for In-store design (in retailing)

The design of a retail store encompasses a multitude of factors, presenting retailers with ample opportunities to shape the shopping environment. As stated by Levy, Weitz, and Grewal (2018), in-store design contributes to hedonic value by fostering an engaging and enjoyable shopping experience. Gestalt's principles clarify how overall design impacts perception and serves as fundamental guidelines for crafting retail spaces of all kinds (Mau, Schweizer, & Oriet, 2023). The primary objective is to develop a retail environment characterised by coherence and clarity, aiming to satisfy customers' desire for meaningful and structured experiences. Factors such as type of customer, store layout and size influence customer traffic patterns within stores (Ebster & Garaus, 2015).

New theories are continually emerging in retailing and in-store design research, contributing to the development of a more enjoyable retail environment. From this, sensory congruence has emerged, explaining the collaboration between the individual's various senses and the environment (Yakhlef, 2015). From a marketing communication and brand strategy standpoint, congruence has been utilised a great deal in diverse contexts (Fleck & Quester, 2007). In essence, sensory congruence occurs when the atmosphere or information aligns with sensory cues, resulting in congruent (or incongruent) sensory experiences that, to some extent, dictate consumer behaviour. While sensory congruence has been applied in sensory marketing, congruency as a concept is widely applied in retail contexts and recognized to be of importance when designing retail atmospheres (Krishna, Elderand & Caldara, 2010; Helme Falk & Hultén, 2017).

The retailer is obligated to ensure reasonable accessibility for individuals with disabilities (Diskrimineringsombudsmannen, 2022), making it essential to consider new parameters alongside segmenting different disabilities to identify their varying needs when designing a service or retail store. Failure to effectively recognise these needs can result in negative changes and ultimately reduce perceived accessibility to the point of customer dissatisfaction (Baker, Stephens & Hill, 2002). As stated by Baker, Stephens, and Hill *"The needs of customers with visual impairments are simple; accommodation means translation of all visual information required to complete an exchange or facilitate an experience. For example, labels, colours, codes, product lists, and price lists may need to be translated."* (2002, pp. 237), Moreover, achieving marketplace accessibility entails addressing both environmental barriers and changing attitudes toward disabled individuals' experiences.



1.1.3 Being visually impaired

Every day, individuals rely on their five senses to gather essential information from their surroundings. This information is crucial for their ability to respond, engage, and adapt (Harleystreetent, n.d.). Through the combined use of all five senses, individuals gain a comprehensive understanding of the world around them whether they are at home, in a store or on the go. However, within society, there are those who lack one of these senses, living to what may be classified as a disability depending on the severity and impact of their impairment (World Health Organization, 2024). Visual impairment, as defined by the World Health Organization (2023), refers to the impact of an eye condition on the visual system and its functions. Visually impaired people (VIP) experience limitations in utilising their vision in daily activities, rendering them distinct when contextualised in various scenarios (National Eye Institute, 2023). Despite this, visual impairment exists along a spectrum, ranging from those with full visual ability to those who are visually impaired and, ultimately, those who are blind.

In Sweden, visual impairment and blindness are categorised into distinct groups, ranging from category 0 for no visual impairment to category 2 for severe visual impairment, and categories 3 through 5 for blindness (Socialstyrelsen, n.d.). An additional category named 9 denotes undetermined or unspecified visual impairment. These categories are differentiated based on varying levels of visual acuity, including uncorrected, presenting, and best-corrected visual acuity (Appendix A & B). There is no comprehensive registration of people with visual impairments in Sweden, which makes it impossible to say accurately how much of the population has vision loss (Kunskapsguiden, 2022). However, roughly 100,000 individuals suffer from vision impairments severe enough to necessitate contact with one of the nation's vision centres, with nearly 450,000 people over the age of 16 acknowledging visual difficulties. Among these, an estimated 10,000 individuals are either blind or have a significant visual impairment.

The response of individuals with visual impairments to various environmental factors such as darkness or different types of light sources is unique to each person (Kunskapsguiden, 2022). This individual variability presents challenges in formulating precise guidelines for designing the retail environment. For VIP and blind people (BP), a tactile writing system called braille enables communication using written information. In Sweden, 1000 people use braille as their primary means of reading and writing (Synskadades riksförbund, 2024a). For VIP, the marketplace holds significance beyond mere task completion and social interaction; it represents an opportunity for autonomy and the enjoyment of independence (Baker, Stephens & Hill, 2002, pp. 230). Additionally, research indicates that supermarkets are the preferred choice for VIP when it comes to grocery shopping (Kostyra et al. 2017). Stores such as local grocery shops, that lack self-service, or farmers' markets were not as favoured among VIP.



1.2 Problem Discussion

1.2.1 Theoretical Problematization

Yu, Tullio-Pow, and Akhtar (2015) highlight the significant challenges that VIP face during the physical shopping experience compared to the visually abled. They emphasise various aspects of the physical retail environment “*from location, parking, entrance, to store layout, fitting rooms, and ambience factors such as lighting, colour, and signage.*” (Yu, Tullio-Pow & Akhtar, 2015, pp. 124). Previous research by Baker (2006) has indicated that VIP in the context of shopping and participating in the marketplace aspire to attain “*consumer normalcy*” and seek personalised shopping experiences that cater to their unique needs (Baker, 2006, pp. 47). The author contends that symbolic and experiential elements conflicting with the notion of social equality in retail spaces may deter visually impaired consumers.

Regarding the grocery shopping preferences of VIP, Kostyra et al. (2017) suggest that while supermarkets are the favoured choice, they fall short in terms of providing assistance through sales staff. The authors argue that although supermarkets offer better accessibility for the visually impaired in terms of store layout and arrangement, they lack adequate support from staff. However, various technological assistive products are available on the market for VIP. Boldu et al. (2020) have identified three crucial factors influencing attitudes toward these technologies. Additionally, usability varies depending on whether the user became visually impaired later in life or was born blind. Despite this, Plikynas et al. (2020) argue that most existing assistive products do not effectively meet the needs of the target group. While some products are better developed and tailored, their unwieldiness results in low use.

Various factors significantly affect product recognition for VIP. Both Kostyra et al. (2017) and Balconi, Acconito, and Angioletti (2022) advocate for permanent product placement on shelves. Balconi, Acconito, and Angioletti (2022) expand on this concept, suggesting that perceptual and sensory characteristics also influence the recognition process. Additionally, in the decision-making process, sensory factors like taste and flavour are not as influential as the familiarity of purchasing groceries from a recognized brand among VIP (Kostyra et al. 2017).

Yakhlef (2015) states that an individual's single sense alone cannot fully capture all impressions and stimuli. The author states that “*When there is harmony between bodily skills and environmental challenges, the body reaches a temporary grip on the world that provides it with a sense of orientation and security.*” (2015, pp. 559). The author argues that existing research on customer experience in environments primarily focuses on physical design, satisfying its findings in the context of cognition. Helme Falk and Hultén (2017) have examined retail settings, specifically visually dominant retail settings and explored the interpretation and role sensory cues have in that space. They identified various sensory cues and their



impact on customers' emotional and purchase behaviour, highlighting the significance of implementing and combining these cues to enhance the experience of the target group. Additionally, Nguyen et al. (2020) propose that future research on congruent customer characteristics should extend to utilitarian sectors like banks and supermarkets, as well as explore congruence characteristics within employee-to-consumer relationships. In conclusion, addressing the multifaceted challenges faced by VIP as consumers in retail environments requires a holistic approach that includes not only physical accessibility but also personalised assistance, effective technological solutions, and an understanding of sensory cues to enhance their overall shopping experience.

1.2.2 Practical Problematization

In recent history, efforts to enhance the accessibility of retail spaces for individuals with disabilities have intensified, spurred by the implementation of laws and acts in numerous countries aimed at improving accommodations for this demographic (United Nations, 2024a; Svenskhandel, 2022; ANED, 2013; ADA, 2009). An ongoing debate comes in the form of how committed societies really are towards making life accessible for people with a disability. In a press release from Synskadades Riksförbund (2024b) they state that Sweden does not fulfil the requirements put in place by FN for people with disabilities. Analysis and reviews done by civil society organisations claim that the government of Sweden is not doing enough for their citizens with various disabilities. The support in today's society has decreased for people with disabilities, including support and accessibility for VIP (Synskadades Riksförbund, 2024b). Companies and organisations are mandated to modify or eliminate design barriers in their retail environment, however, simply being informed about legal obligations does not necessarily equate understanding of customers with disabilities perception and experience in retail spaces. Supermarkets have a bad reputation among visually impaired customers due to a lack of various functional aids, which results in challenges for VIP who feel that companies struggle to implement adequate assistance beyond the minimum legal requirements (P4 Stockholm, 2014). In Sweden in 2025, a new law takes effect which demands accessibility on a range of products and services to implement and follow the European Parliament's directives on accessibility (SFS 2023:254). The law is comprehensive, providing demands on the presentation of information for the purpose of perceiving information effortlessly and with more than one sense (Socialdepartementet, 2022). In supermarkets, these demands mainly affect payment functions such as card terminals and self-checkout terminals in-store.

An illustration of a company within the food retailing sector addressing this challenge is ICA. They have implemented a queuing system capable of audibly announcing the next number for assistance (ICA, 2014). This system was developed with VIP in mind and was introduced following consumer requests for improved accessibility aids. Prior to this initiative, the company had not integrated any technological aids to ensure the retail experience was as accessible for VIP as



it was for sighted consumers. In addition, markings in braille on products are only used on prescription medicines, making it impossible for the visually impaired to distinguish products such as dairy products and canned tomatoes from each other (Synskadades riksförbund, 2024a). Although instances of VIP being denied access to supermarkets are not widespread, they do occur. For example, one company refused entry to a visually impaired customer accompanied by a service dog, which is essential for the customer's daily activities (P4 Jämtland, 2010). Examining this research could provide companies and organisations with a deeper understanding of the challenges faced by VIP during food shopping in physical retail spaces. By researching VIPs' retail experiences and analysing them from a perspective of sensory congruence, in-store design, and the various aids they utilise, supermarkets and other retail stores could significantly enhance their store designs to better accommodate visually impaired customers.

1.2.3 The Research Gap

During the research of this thesis, it became evident that the topic of VIP as consumers is not only significant to study but also lacking in consideration and existing research. For instance, Kostyra et al. (2017) highlight the limited amount of research on visually impaired individuals as consumers, particularly concerning their experiences with food shopping. The authors argue that future research ambitions are essential for gaining a deeper understanding of the unique needs, expectations, and desires of blind individuals in the realm of grocery shopping. Similarly, Shockley, Roth, and Fredendall (2011) advocate for further research to uncover the obstacles faced by visually impaired consumers during food shopping. The authors emphasise the importance of investigating how store layout, signage, aisle width, displays, and other factors impact shoppers' information processing and capacity to engage with the given product mix effectively. Given that the sense of orientation and security relies on the alignment between bodily capabilities and the environment (Yakhlef, 2015), there is a call for future research to explore how sensory characteristics of the environment can be tailored to foster harmony for customers with diverse disabilities.

Numerous researchers advocate for a deeper understanding of how VIP experience the retail environment. For instance, Imschloss and Kuehn (2017) state the necessity for future studies examining the impact of retail store atmospherics on consumer behaviour concerning various aspects of multisensory interactions. They also emphasise the importance of investigating how factors such as product selection are influenced by multisensory congruence. Similarly, Helmefalk and Hultén (2017) identify a research gap from both theoretical and managerial perspectives, suggesting that retailers should view multisensory congruent cues as experimental. They propose further investigations to determine the most appropriate multisensory cues for specific retail stores, products, services, and brand images. Previous research has predominantly focused on sighted consumers, analysing how sensory cues are interpreted in retail settings. However, there is a noticeable lack of research examining how the absence or distortion of one sense



affects the retail experience. This highlights the relevance of conducting research on this subject, particularly concerning how VIP experience retail spaces in supermarkets. VIP constitute a significant customer segment deserving of equitable accessibility and retail experiences (Dodds & Palakshappa, 2022). Addressing this requires filling the research gap to gain insights into their unique experiences and needs within retail environments.

1.5 Purpose

The purpose of this study is to explore how people who are visually impaired experience the retail setting of supermarkets. The study aims to analyse how in-store design and sensory cues can be used to enable and affect supermarkets for the visually impaired. Additionally, explore the impact of using aids to facilitate the retail experience.

1.4 Research questions

RQ1: How do supermarkets in-store design and sensory cues affect the visually impaired's retail experience?

RQ2: How do aids contribute to the retail experience for visually impaired people in supermarkets?

1.5 Delimitations

This thesis is limited to the VIP in Sweden, specifically in the supermarket retail space. The research is conducted towards the B2C market in retail business, orientated from a customer perspective, in this case, the VIP customer base in order to create an understanding of how to facilitate the retail experience for visually impaired customers in supermarkets. Therefore, the perspectives of other customer bases such as sighted people are not considered. VIP do consider supermarkets as the preferred choice when it comes to grocery shopping (Kostyra et al. 2017), supporting the research context of VIP in that environment. The research will also focus on supermarkets which mainly distribute food-related products. Through this paper, visually impaired people and blind people will be referred to as VIP and BP. BP are included in VIP, therefore if BP is referred to on their own it excludes other types of visual impairments.

2. Literature Review

During this chapter, the literature review which is the foundation for this thesis is presented. The chapter begins with explaining the use of senses in retail and the effect on both sighted and visually impaired people, followed by different aspects of the retail experience in Supermarkets for visually impaired people. The chapter



is summarised with a discussion of how it relates to the identified research gap, ending with a conceptual framework providing a visualisation of the theories for the reader.

2.1 The use of senses in retail

In the field of customer experience, perception is a concept that combines sensory inputs and environmental factors, as noted by Yakhlef (2015). The author argues that the sensations perceived by individuals play a crucial role in shaping the retail environment, arising from an ongoing interaction between the individual and their surroundings. Through the utilisation of the body and its various senses, one can foster orientations and feelings of security, emphasising the importance of adopting a multisensory perspective. Krishna and Morrin (2008) state that individual interpretations of nondiagnostic haptic cues vary among consumers. These cues, which are sensory inputs unrelated to the immediate task, contribute to the overall perception. In Altay's (2021) research it was discussed that in retail environments, it is crucial to include diverse individuals and build upon the accessibility through the various senses. Yakhlef (2015) supports this statement by suggesting that no single sense comprehensively captures the entirety of an experience. Customers derive a sense of security and orientation when all their physical abilities harmonise with the challenges posed by the environment. This interaction involves factors such as the surroundings, one's own body, objects, and other individuals. Similarly, Krishna and Morrin (2008) search into the perceived implications of haptic feedback and touch concerning product attributes and value within the context of nondiagnostic haptic cues, supporting the view of perception from different sensory cues. Eklund and Helmfalk (2022) provide an additional perspective to the discussion of the use of senses, arguing that sensory and semantic cues intertwine, with sensory cues complementing semantic meaning, particularly impacting product design and atmosphere.

Helmfalk and Hultén (2017) conducted research regarding the impact of multi-sensory congruent cues within a retail store atmosphere on shoppers' emotions and purchase behaviour. Cues encompassed auditory and olfactory stimuli alongside strategically positioned visual cues. The study highlighted the importance of motivators in pinpointing essential factors for individual retail environments, guiding the prioritisation and application of sensory cues. The authors argue that auditory and olfactory cues are effective in enhancing visually dense atmospheres. This follows Yakhlef's (2015) statement about the limitations of only using singular senses. However, the addition of extra visual cues to an already visually dense environment did not have positive effects on shoppers' emotions or purchase behaviour (Helmfalk and Hultén, 2017). Even when the added visual cue aligned with the existing atmosphere, no discernible positive impact was observed. This phenomenon aligns with Ekelund and Helmfalk's (2022) theory of non-intrusive modification and cue pairing, emphasising the importance of subtle adjustments in incongruence and understanding the trade-offs in cue combinations. Further,



Helmefalk and Hultén (2017) advocate for the consideration of non-visual cues like auditory and olfactory stimuli as complements to existing visual cues for long-term customer satisfaction and store image enhancement. Additionally, the authors argue that the relevance of each sensory cue depends on factors such as customer preferences and the nature of the product category and its messaging. However, the study had limitations, notably the absence of sensory cues such as touch. Thus, future research should explore the implications of integrating tactile congruent cues and their potential impact on shoppers' emotions and purchase behaviour. Furthermore, Hultén (2012) states that the sensory cue of touch was useful strategically when enhancing consumer's need for touch. Consumers sometimes need the complementary evaluation of touch when finalising their purchase behaviour.

Sandell (2019) conducted a study of the relationship between the consumer's characteristics and the effectiveness of the olfactory cues on purchase behaviour. Olfactory in itself cannot be switched off and its cues can affect consumers differently. Their findings showed that olfactory cues can positively influence consumer purchase behaviour. This aligns with Helmefalk and Hultén's (2017) previous claim about olfactory cues effectiveness on purchase behaviour. Essentially, it depends on the consumer's holistic values and the gender combination (Sandell, 2019). Conclusively, sensory cues have a direct link to actual behaviour that goes beyond attitudes and memories. This also can be linked partially to Yakhlef's (2015) claim about customers' perceptions and experiences. Sandell (2019) further claims through the research that olfactory cues have a potential use and should not be limited to considering age nor gender, but other factors of customers.

Eklund and Helmefalk (2022) explore the complex relationship between congruence and incongruence, particularly regarding products, brand identity, and atmosphere. Through a theoretical framework, the authors analyse how congruence manifests in various contexts, incorporating elements such as time, space, and optimal engagement. However, the findings suggest that establishing conclusive evidence regarding the permanence or primary influence of congruence is challenging due to the diverse contexts, cues, and theoretical perspectives involved. Nevertheless, the authors state that congruence and incongruence does not always manifest as anticipated. Consistent development of positive feelings relies heavily on recognizing congruence between product, brand, and atmosphere. Incongruence, on the other hand, can initially capture attention and effectively differentiate, but it is crucial to acknowledge consumers' perceptions of space and time as congruence evolves continuously. Understanding consumers' emotional and behavioural expectations, intensity, and cognitive processes such as memory and schemas is vital. Managing incongruence non-intrusively and navigating the trade-offs in pairing cues are essential. On the other hand, Heckler and Childers (1992) define congruence as the combination of relevance and expectancy, distinguishing between congruent information that is either expected-relevant or unexpected-relevant. Eklund and Helmefalk (2022) state that different consumers



may perceive paired cues differently, recalling the necessity of understanding the customer perspective to adapt atmosphere, cues, and timing for a positive outcome. This research, if applied in the context of VIP retail experiences, could offer valuable insights for retailers seeking to enhance accessibility and foster stronger customer relationships. Furthermore, the authors emphasise the consumer's initial role in the ongoing process of achieving congruence which suggests that understanding the consumer's evolving perception is crucial in maintaining congruence over time.

A related cause that might occur when conducting using senses, specifically multisensory is the consequence of sensory overload mentioned by Spence et al. (2014). It was found that in principle, it is a good idea to introduce multiple senses to expose the consumer. This aligns with other literature on the perception of sensory cues (Yakhlef, 2015; Eklund & Helmefalk, 2022; Sandell, 2019), attaining favorability in both attitude and behaviour (Spence et al. 2014). However, increasing sensory cues has the consequence of affecting the sensory perception points too much. This can ultimately lead to the risk of sensory overload in itself. Each person has their own preference of amount of sensory stimulation so there is not a clear answer, however, the authors did find that sensory cues that do not align with each other can see results depending on the level of exposure where the combined signals lead to less enjoyable for the customer. Similar to what sensory congruence is defined as (Heckler & Childers, 1992; Helmefalk & Hultén, 2017; Eklund & Helmefalk, 2022). It was also found that two atmospheric stimuli achieved positive outcomes, but three congruent stimuli inherently added negative effects (Spence et al. 2014). Finally, depending on the level of sensory dominance, sensory cues can coexist but with consideration of intensity. This is depending on the level of stimuli. Eklund and Helmefalk (2022) also stated in similar ways the importance of optimal engagement to successfully manage the customer through congruence. Studies about understanding multisensory atmospheres are needed to understand the overall perspective that sensory cues can impact consumer's behaviour and experiences, this should be seen as complementary rather than merged as one singular aspect (Spence et al. 2014). More research is needed in this area because of the difficulties in studying the implications surrounding it.

In contrast to the perspectives of Helmefalk and Hultén (2017) and Ekelund and Helmefalk (2022), Nguyen et al. (2020) introduce a new viewpoint by incorporating additional factors into the discussion of congruence. Nguyen et al. (2020) highlight the absence of studies focusing on customer-to-customer interactions within the context of congruence and how customer characteristics influence service experiences. This gap in understanding means that many managers are unaware of the role of customer characteristics in shaping service encounters and how different customers perceive one another in such situations. The lack of knowledge regarding the impact of specific customer attributes can lead to negative consequences for the overall customer experience, particularly affecting VIP (Kostyra et al. 2017; Celik & Yakut, 2021). Nguyen et al. (2020)



argue that congruence surrounding customer characteristics significantly contributes to the overall service experience. Their findings suggest that congruence in customer characteristics plays a vital role in determining key positive outcomes such as satisfaction in service contexts. Furthermore, the research indicates that incongruent customer characteristics carries greater influence than congruent ones, particularly concerning focal customer outcomes. In high-involvement consumption scenarios, congruence in customer characteristics positively impacts focal customer satisfaction and purchase intention, while this effect is less pronounced in low-involvement scenarios where congruence in customer characteristics is deemed less critical.

The overarching conclusion drawn by Nguyen et al. (2020) is that congruence in customer characteristics exhibits variability based on the level of involvement within a retail setting. Their research reveals a correlation between heightened status enhancement and disadvantage effects of incongruent customer characteristics on focal customer outcomes. Furthermore, the authors introduce customer mood as a new critical factor influencing the relationship between congruence and customer characteristics, thereby extending the conceptual framework established by Helme Falk and Hultén (2017) and Ekelund and Helme Falk (2022) in retail environments. Nguyen et al. (2020) suggest that a managerial review of their findings can offer valuable insights, particularly in industries like food service, retail and broader service sectors, where interactions range from high to low involvement. This recommendation stems from the perspective of maintaining and enhancing brand image through effective engagement with the customer base. Frontline service employees should bear in mind that understanding, representing, and leveraging congruent customer characteristics can foster positive engagement during service encounters. This aligns with a discussion on the importance of personnel assistance for VIP, where personnel interactions have proven crucial for tasks such as in-store navigation, product recognition, and overall experience for customers in Supermarkets (Guo & Bigham, 2018; Yu, Tullio-Pow & Akhtar, 2015; Kostyra et al. 2017; Celik & Yakut, 2021) which will be discussed later on in the literature review.

2.2 VIP - In-store designs impact

Today, when examining accessibility, companies separate customers with disabilities from the rest instead of seeing accessibility as a main factor for all (Dodds & Palakshappa, 2022). This approach needs to be changed in order to create an inclusive retail environment, acknowledge accessibility as a factor in the centre of design. In addition, understanding different disabilities and identities as well as how to facilitate in order to provide service inclusion is crucial, especially if the company strives to create an improved environment. In a study conducted by Balconi, Acconito, and Angioletti (2022), the implicit emotional consumer experience of VIP while in-store was explored, revealing significant obstacles encountered during food shopping. Data encompassing autonomic, behavioural,



and self-reported measures were collected across three stages of the shopping experience: product identification, style of product purchase, and overall consumer experience. The study involved two distinct groups: VIP and a control group. Findings demonstrated that VIP had “*higher levels of disorientation, difficulty in finding products, and repeating the route independently*” (Balconi, Acconito, & Angioletti, 2022, pp. 1). Additionally, comparisons between the control and VIP groups revealed that VIP experienced heightened feelings of stress and frustration, alongside greater cognitive exertion. However, despite VIP exhibiting less accuracy in product recognition, their ability to identify and search for specific items did not significantly differ from the control group. The authors state that various perceptual and sensory characteristics, as well as product placement, influence the recognition process, rendering certain products, such as fruits, easier to identify than others, like frozen foods. To enhance the emotional customer experience for VIP, the authors recommend implementing in-store marketing strategies such as braille maps, tactile touchpoints and guided exploration of the store upon initial entry.

Orientation and Customer Traffic in-store are two factors which need to be managed. Ebster and Garaus (2015) provides several overarching principles govern Customer Traffic within stores: the transition zone immediately after entry is crucial for customers to acclimate to the store environment, although it may initially hinder perception; narrow aisles are often avoided due to the discomfort of invading on personal space; and shoppers tend to favour the entrance floor over upper and lower levels. Contrary to the findings of Balconi, Acconito, and Angioletti’s (2022), Kostyra et al. (2017) contend that VIP encounter greater challenges with product orientation compared to sighted customers. The authors describe various factors indicating VIPs' preference for stores to facilitate easier shopping experiences, thus corroborating the notion that VIP may indeed experience discomfort while grocery shopping, as suggested by Balconi, Acconito, and Angioletti (2022). Kostyra et al. (2017) argue that retail establishments possess the capability to assist blind and visually impaired customers in locating items by establishing permanent placements for products on shelves or minimising changes to product positioning. But according to Drèze, Hoch and Purk (1994), Bianchi et al. (2021) and Valenzuela, Raghurir, and Mitakakis (2013), it was found that there is an overall profit for both the retailer and manufacturer via the considerations of shelf management. This makes it appealing for retailers to, for example, consider temporary and permanent display characteristics (Drèze, Hoch, & Purk, 1994). Temporary displays are effective in attracting attention from the consumer but are intrusive because of their large size and novelty. On the other hand, permanent displays are also effective in capturing attention with their constant, but dynamic manipulation, for example, product placement, product adjacencies and aesthetic elements.

This assertion is further elaborated upon by Mau, Schweizer and Oriet (2023), who emphasise that the placement of products on shelves directly impacts product



orientation, suggesting avoidance of items being placed too high. The establishment of a coherent mental map for customers and their memory facilitates a smoothly functioning cognitive framework of the environment, thus enabling customers to navigate effortlessly within it. According to the authors, when concerning customer satisfaction and shaping the shopping experience, a well-defined structure and organised shelves exert significant influence. Retailers often prioritise grouping products within the same category and providing assistance to swiftly locate desired items. Furthermore, Kostyra et al. (2017) propose that stores considering product placement and employing large, printed handouts or high-contrast signage could significantly enhance shopping convenience and accessibility for visually impaired individuals, thereby promoting their independence. Additionally, a majority of VIP express a preference for larger fonts on product labels.

According to Kostyra et al. (2017), VIP prioritise price and brand over sensory aspects like taste and flavour when selecting food products, unlike sighted customers. The decision-making process for VIP tends to revolve around purchasing groceries from familiar brands rather than focusing on sensory attributes. The authors state that VIP exhibit brand loyalty due to limited access to information about new brands and products, therefore brand loyalty can be seen as forced loyalty. This assertion aligns with the findings of Guo and Bigham (2018), who suggest that BP primarily concentrate on the functionalities of familiar items . Kostyra et al. (2017) also reveal that visually impaired consumers tend to avoid a wide variety of food products. The reason for this, stated by Kostyra et al. is: *“may be due to many obstacles they face when it comes to evaluation of external quality cues and feelings of discomfort associated with asking others for assistance”* (2017, pp. 21). These insights provide additional perspectives on the earlier-discussed topic of product orientation for VIP and offer a deeper understanding of their product choices.

Yu, Tullio-Pow, and Akhtar (2015) advocate for a store layout that prioritises clarity and simplicity to better accommodate the mobility challenges faced by visually impaired individuals. According to the study, the majority of visually impaired participants experienced stress when navigating through stores with unclearly labelled departments, corroborating the argument put forth by Kostyra et al. (2017) regarding the importance of enhancing VIPs' independence in-store. Yu, Tullio-Pow and Akhtar (2015) assert that VIP are at risk of inadvertently purchasing items from the wrong sections or departments due to poor organisation, providing further perspectives on the discussion surrounding product orientation for VIP. The authors propose that distinguishing walkways from merchandise areas would ease navigation challenges for VIP. For instance, using carpets at shelves and tiles in walkways would provide directional cues and aid in differentiating store areas. However, it's crucial for retail stores to carefully consider the implementation of elements like carpets, as highlighted by one respondent in Yu, Tullio-Pow, and Akhtar's (2015) article, who mentioned the potential tripping



hazards posed by thick metal edges on carpets. Additionally, VIP face obstacles in freely moving around retail spaces, with narrow aisles and protruding display fixtures outside shelves designed areas posing safety concerns (Yu, Tullio-Pow & Akhtar, 2015; Mau, Schweizer & Oriet, 2023). In line with this, Celik and Yakut (2021) argue that incorporating guide roads on the floor can enhance safety, comfort, and self-confidence for VIP by indicating obstacle-free pathways.

According to Mau, Schweizer, and Oriet (2023), enhancing the overall orientation for all customers can be achieved using large lettering, images, colours, as well as wall and top signs. Conversely, Yu, Tullio-Pow, and Akhtar (2015) suggest that easily navigable physical retail space and signage composed of sizable fonts and non-cursive letters contribute to more accessibility. Additionally, the importance of colour contrast on signage is highlighted, with clear distinctions between letters and background enhancing readability - a viewpoint that aligns with the theory proposed by Kostyra et al. (2017) on the impact of high-contrast signs on accessibility for VIP. However, Baker, Stephens, and Hill (2002) argue that retailers should not only consider fonts, sizes, and colours but also pay attention to the placement of signage. Signs positioned directly over the middle of grocery aisles are often challenging to read, a factor which has been identified as crucial for the behaviour of exclusively shopping at one specific store only. While Yu, Tullio-Pow, and Akhtar (2015) advocate for the use of fluorescent lighting as environmental lighting, the authors caution against its use on signage, particularly when it involves flashing digital lights. When utilised effectively, signage that clearly indicates the location of specific merchandise on shelves can be beneficial for VIP, thus enhancing the retail store experience with signage as a way-finding tool. Quartier, Christiaans, and Van Cleempoel note an interesting point about lighting, arguing that *“consumers are not aware of differences in lighting”* (2008, pp. 216/12). Lighting can be difficult to assess and see differences between. However, no conclusion could be made about its effects on consumers' mood and behaviour.

Both Baker, Stephens, and Hill (2002) and Yu, Tullio-Pow, and Akhtar (2015) support the relevance of Mau, Schweizer, and Oriet's (2023) orientation theory, but emphasising the need for adaptation in the context of VIP to yield positive effects. However, lighting adaptation highlights a challenging factor for VIP shoppers (Yu, Tullio-Pow & Akhtar, 2015), suggesting that store lighting significantly influences their experience, particularly concerning the contrast between various light levels, such as bright and dark areas. As a result, ensuring the right lighting in retail stores is essential for the satisfaction of VIPs' experience. The authors argue that VIP prefers fluorescent lighting that is not excessively bright and remains consistent throughout the store. Moreover, store lighting can facilitate colour contrast between shelves, floors, and walls, aiding VIP in navigation. In conclusion, enhancing colour contrast from general lighting to various surfaces could potentially ease navigation for VIP, offering a new perspective on in-store orientation and design.



In today's society, physical interfaces are becoming increasingly prevalent. Guo and Bigham (2018) highlight that for BP, these interfaces are generally inaccessible. Typically, BP relies on tactile markings, self-voicing interfaces, or sighted assistance in their daily activities. However, tactile markings may not always be available on public interfaces, which can inherently delimitate interaction. For instance, semi-structured interviews conducted by the authors revealed that everyday interfaces are becoming less accessible. Self-checkout terminals at grocery stores were cited as examples that diminish the independence for BP. Guo and Bigham (2018) argue that the transition from physical buttons to flat touch pads poses challenges for those who rely on memorization to operate devices, thus reducing accessibility. To address these issues, the authors identified four design requirements for creating systems that meet the accessibility needs of users, particularly for home appliances. These requirements include accommodating BP's preferences for labelling home appliances, advocating for diverse labelling strategies and providing attentive tactile feedback. Additionally, recognizing the difficulty BP faces in memorising abbreviations and functions of appliances is essential for enhancing accessibility.

According to Guo and Bigham (2018), VIP tend to utilise only familiar functions, which limits their ability and potential to learn. Moreover, the authors argue that appliances such as self-checkout terminals, with dynamic interfaces, necessitate dynamic design solutions. Simply applying tactile markings may not suffice to address the screen changes that occur during interaction. Yu, Tullio-Pow, and Akhtar (2015) highlight the stressful nature of the checkout process for VIP during the retail experience. Retailers can improve the checkout process for visually impaired consumers by different methods, such as: *“the use of receipts with a large font and a bold signature line; the proper handling of change by the cashier; the provision of user-friendly debit card machines for the visually impaired; and possibly an audio-enhanced checkout system that announces the price and quantity of merchandise being scanned.”* (Yu, Tullio-Pow & Akhtar, 2015, pp. 127). This reinforces Guo and Bigham's (2018) assertion regarding the need for diverse design solutions to enhance the in-store retail experience and develop accessible physical interfaces. However, the notion that tactile markings alone are insufficient contradicts Balconi, Acconito, and Angioletti's (2022) reasoning regarding the implementation of strategies involving tactile touchpoints to enhance the emotional customer experience for VIP. On the other hand, Kostyra et al. (2017) report that among VIP, 84% found products labelled in braille helpful, with nearly all respondents expressing a desire for additional information, such as product name, price, and expiration date, also available in braille. However, Slade and Edwards (2016) state that only 8% of VIP in the UK are proficient in reading braille and as mentioned earlier in the paper, in Sweden there are 1000 people using braille as their primary means of reading and writing (Synskadades riksförbund, 2024a). This lack of proficiency could undermine the general utility of braille as a helpful factor for all VIP.



2.3 Aids for VIP - technological and personnel aid in-store

2.3.1 Technological aids

Numerous research has been previously conducted to aid VIP in their daily lives, resulting in the development of various assistive technologies tailored to different situations and needs. Boldu et al. (2020) have pinpointed three crucial factors regarding the perception and acceptance of technological assistive products: extended functionalities, recognizing that VIP do not require constant assistance and may not wear a device all day; accuracy on the task, emphasising the importance of specific and precise information for VIP; discreetness, highlighting the desire among VIP to avoid being labelled as disabled, particularly for those experiencing gradual sight loss over time. The study revealed that products aiding visual interpretation are most effective for individuals who have lost their sight later in life, as they can leverage prior experience to locate labels and other information. Additionally, audio explanations from these devices were found to be more effective in quiet environments. However, Kostyra et al. (2017) reported in their article that the majority of visually impaired and blind consumers found electronic readers to be helpful. In fact, 72% of respondents in the article indicated that devices capable of reading labels facilitated their food shopping experience, enabling them to access information on product labels.

Plikynas et al. (2020) conducted a comprehensive overview of indoor navigation systems, providing a qualitative assessment of these systems. The overview primarily focused on non-widely available commercial solutions, and assessments were based on criteria suggested by blind experts. The review reveals that the majority of assistive systems either remain in the active development stage or exist solely as prototypes, a finding that aligns with the three critical points identified by Boldu et al. (2020). Plikynas et al. (2020) argue that the scarcity of fully developed products, coupled with existing commercially available assistive devices failing to meet the needs and expectations of VIP, as well as VIPs' hesitant to trust new equipment, underscores the necessity for further development in this field. While smartphones may offer a potential solution in the future (Plikynas et al. 2020; Celik & Yakut, 2021), current models lack the necessary computational speed for high-flow real-time image processing (Plikynas et al. 2020). Celik and Yakut (2021) suggest that retailers should create a multisensory environment to assist VIP in navigation, utilising the store's own mobile application and audible technologies. Handheld scanners are a common shopping aid in supermarkets. Yu, Tullio-Pow, and Akhtar (2015) suggest that for VIP, it would be particularly beneficial if these scanners could identify tags and signs through audio cues and read-aloud functionalities.

For VIP when it comes to navigating their surroundings, Plikynas et al. (2020) observed that while obstacles within a 2-metre range can be easily identified with a cane, VIP require information about significant environmental elements within a



range of 5-20 metres. However, VIP also express a need to receive information about significant environmental elements within a range of 5-20 metres. Many non-camera-based orientation technologies are vulnerable to various factors depending on the circumstances, potentially hindering their mobility, flexibility, and universality, or necessitating additional monitoring, thus rendering them problematic and often avoided. Moreover, orientation equipment often relies on an internet connection to utilise a map of navigational tags. In the analysis of vision-based indoor navigation, Plikynas et al. (2020) observed a growing preference for 3D cameras equipped with RGB-D sensors, which can detect objects, estimate distances and distinguish various levels of objects, thereby facilitating navigation for VIP. Additionally, there is ongoing development of smaller cameras; however, their size and unwieldiness are perceived obstacles to their utilisation. Hybrid indoor navigation, a novel product category, combines different technologies to enhance system accuracy, although it necessitates more infrastructure elements compared to other vision systems.

2.3.2 Personnel aid

Even though these technologies are designed to facilitate everyday life, Boldu et al. (2020) have identified a low frequency of use of these technologies. The reason is that VIP find it easier to ask others for help since the assistive products are complicated, unmanageable and not adapted to VIP needs (Boldu et al. 2020; Plikynas et al. 2020). In addition, Guo and Bigham (2018) emphasise that accessibility for the blind often necessitates sighted assistance. According to Yu, Tullio-Pow and Akhtar (2015) retail sales associates need to be able to assist customers who are visually impaired by having knowledge about the store's product and how to meet the needs of this customer segment. The retail experience includes the interaction with sales associates and is a critical factor for visually impaired customers. Therefore, discussing in-store personnel aid is highly relevant. Kostyra et al. (2017) report the aspect that facilitates the shopping experience the most for visually impaired consumers is the helpfulness of sales staff with 99.2% of their respondents mentioning it as an enabling factor for food shopping, which reinforces Boldu et al.'s (2020) findings. However, Kostyra et al. states that VIP can experience "*feelings of discomfort associated with asking others for assistance*" (2017, pp. 21), which argues for an obstacle for VIP to take help, resulting in them not being able to take part in the whole experience in an optimal way. As earlier mentioned, VIP prefer supermarkets over local grocery shops and farmers markets. This is an interesting choice since Kostyra et al. (2017) have found that supermarkets are at a disadvantage when it comes to the lack of assistance they are able to provide from the sales staff compared to the other alternatives. Therefore, implementing a customer-focused service approach and providing employee training for interacting with customers with disabilities are two strategies that significantly enhance the customer experience (Baker, Stephens & Hill, 2002).



Similar to Kostyra et al.'s (2017) findings, Celik and Yakut (2021) state that staff interaction with customers is a decisive factor in VIP retail experience. Celik and Yakut (2021) studied the relationship between customer satisfaction and perceived vulnerability as well as the relationship between customer satisfaction and repurchase for VIP. The result showed that satisfaction with the store and its service increased as the perceived vulnerability increased, a phenomenon where the staff's interaction played a crucial role. This is explained by Terblanche (2018) since staff interaction is one of the three factors which have been proven to be of particular importance for customer satisfaction as it comes to in-store Customer Shopping Experience. Celik and Yakut (2021) argue that factors that impact blind customers' feeling of social inclusion and acceptance in a specific retail store are service quality, store accessibility and store ambience. The authors state that the analysed relationships do not differ for consumers who are born blind and for those who have become blind over time. Therefore, Celik and Yakut (2021) argue that customer service aimed to satisfy all customers without any discrimination is preferable and can also be a part of a differentiating business strategy. This argument is strengthened by Doods and Palakshappa (2022), however, the authors state that retailers today lack both skills and knowledge to facilitate and provide services to customers with disabilities. Despite this, the personnel possess the possibility of impacting the customers everyday life, enhancing both the experiences and the overall well-being.

In Taylor et al.'s (2019) study, retailer customer communication was shown to *"either construct a disabling environment or facilitate communication access for people with disabilities"* (Taylor et al. 2019, pp. 233). Factors such as ignorance, patronising or stereotyping were identified barriers, negatively impacting the communication. Changes to facilitate the shopping experience for customer with disabilities occurs mainly due to changes and accommodations on the customer's side, rather than improvement made by the retailer. The authors state the need for future research to identify and prevent communication barriers as well as factors for developing an inclusive environment. Guo and Bigham (2018) state that BP generally rely either on tactile markings, self-voicing interfaces or sighted assistance in everyday life. However, tactile markings cannot always be on public interfaces and inherently make it more difficult to interact with. In other words, retailers cannot rely on physical aids only, making a well-functioning system with both physical and personnel aids as well as creating favourable conditions for the use of technical aids necessary.

2.4 Summary

According to research findings, customising and designing in-store elements are crucial for enhancing the retail experience of people with visual impairments. This entails creating accessibility, facilitating product orientation and recognition, and effectively implementing aids within the store. Additionally, creating a retail environment that supports visually impaired individuals in accomplishing daily



tasks and experiencing a sense of autonomy can lead to greater independence (Baker, Stephens & Hill, 2002; Kostyra et al. 2017; Guo & Bigham, 2018). Theories on design elements vary depending on whether the main target audience is sighted or visually impaired customers. However, all theories concur that using said elements as orientation tools is advantageous (Yu, Tullio-Pow & Akhtar, 2015; Baker, Stephens & Hill, 2002; Mau, Schweizer & Orient, 2023). Nevertheless, adapting these tools is necessary specifically for visually impaired customers (Yu, Tullio-Pow & Akhtar, 2015; Mau, Schweizer & Orient, 2023; Baker, Stephens & Hill, 2002).

Furthermore, both Plikynas et al. (2020) and Boldu et al. (2020) contend that existing technological aids for VIP are cumbersome and fail to meet the necessary criteria for effective functionality. Concurrently, interfaces are becoming increasingly inaccessible for VIP (Guo & Bigham, 2018; Plikynas et al. 2020). Despite these challenges, hybrid technological aids are progressively improving in accuracy and level of advancement. Additionally, Boldu et al. (2020) observed that the efficacy of aids varies depending on whether the user is congenitally blind or acquired blindness over time. Looking forward, the significance of human assistance is paramount. Creating a well-functioning system for VIP necessitates a combination of physical aids and personnel support. This is particularly crucial considering that the majority of customers often require sighted assistance (Guo & Bigham, 2018) and find it more convenient to seek help from store staff (Boldu et al. 2020; Plikynas et al. 2020), who also enhance the shopping experience (Kostyra et al. 2017).

Customer retail experience is an abundant experience of cues, inputs, perception and senses (Yakhlef, 2015). Eklund and Helmeffalk (2022) emphasise the importance of recognizing the consumer's initial role and perspective concerning space and time, as congruence is influenced by an ongoing process. However, consumers themselves are different and call for different needs of sensory cues (Altay, 2021; Krishna & Morrin, 2008). Successful utilisation of sensory cues can positively impact customers' emotions and purchasing behaviour (Heckler & Childers, 1992; Helmeffalk & Hultén, 2017; Hultén, 2012; Sandell, 2019). But unsuccessfully, sensory cues can be seen as a hindrance to customers' overall experience (Spence et al. 2014). Establishing multi-sensory congruency in the retail environment (Eklund & Helmeffalk, 2022; Heckler & Childers, 1992; Helmeffalk & Hultén, 2017), along with acknowledging and involving the customer's role in the continuous congruency process, holds significant importance (Eklund & Helmeffalk, 2022; Nguyen et al. 2020). Lastly, to achieve this desirable congruence, the importance of customer and staff interaction should not be overlooked.

The existing research on the retail experience for VIP, including in-store design, aids, and senses has been conducted separately, lacking both practical and theoretical correlation between the themes. While sensory cues play a significant



role in the in-store design, it tends to be overlooked in terms of how VIP perceive and experience it. Additionally, sensory cues may vary significantly for visually impaired consumers, particularly when their shopping experience is influenced by both personnel and technological aids. Although each of these themes presents intriguing topics on its own, an evident research gap emerges when they are combined and analysed together. To gain a deeper understanding of the shopping experience from the perspective of this specific customer segment, further research is warranted.

2.5 Conceptual framework

Following the literature review, crucial concepts can be identified which are essential for exploring the retail experience within supermarkets. The concepts work as cornerstones for shaping a conceptual framework that visualises existing literature. The illustration (see Figure 1) showcases the three dimensions of the retail experience VIP encounter during their supermarket visits. The literature review presents theories and research surrounding the senses, in-store design and aids in the retail setting which combine works together to create one singular experience. The literature depicts the areas as three individual concepts which act like building blocks for the retail experience in the setting of supermarkets. The blocks co-exist according to the literature but do not affect or impact each other in any direct way. From the literature review customers who are visually impaired encounter the retail experience whilst in a retail setting, the setting which in this study is defined as supermarkets. The retail experience and VIP are connected to each other but not necessarily a part of one another making them two separate entities.

3. Methodology

In this chapter, the different methodologies, approaches and strategies used in this thesis are introduced and explained by the authors to provide an understanding of the conduction of the study. Through the chapter, why each chosen methodology is best suited and relevant for the research purpose will be explained. Finally, the authors end the chapter with a discussion of both ethical and sustainable considerations as well as the quality of the research.

3.1 Research Approach

One of the most crucial aspects of understanding the theory and its objectives is to understand the relationship with its phenomenon and the emerging theories and approaches surrounding it (Saunders, Lewis & Thornhill, 2023; Bell, Bryman & Harley, 2019). These are concepts that are commonly used to explain that bond in this kind of research. In this study, the theory of discipline was originally derived from a deductive approach, in which the theory is the foundation and comes before



the data (Rashid et al. 2019). This was to strengthen the initial academic foundation of the work. The literature review, specifically the peer-reviewed articles will be the key to exploring the unknown in this context. However, an abductive approach was favoured and implemented in its stead because of research limitations on VIP in this environment (supermarkets). The abductive approach is a concept where a known premise is a key motivator for the researcher to apply testable conclusions (Saunders, Lewis & Thornhill, 2023). It weaves between the theory of deduction and the data-driven induction for its own space.

Rashid et al. (2019) further explains that abduction does admit prior theories, which only increases the theoretical foundation, unlike induction. It is also considered less theory-guided, unlike deduction, which makes it more flexible. Bell, Bryman, and Harley conceptualised the concept as “*abduction is proposed as a way of overcoming the limitations associated with deductive and inductive positions*” (2019, pp. 24). The hypothesis is susceptible to change because of the tested data (Saunders, Lewis & Thornhill, 2023). A back-forth relationship between the data and the theory is prompted. Because of the obtained empirical data, the theory will be able to evolve and will easily delve deeper into aspects that matter. This will also reflect on how the data collection is processed and used. Those who participate in the research attain their own views, beliefs and experiences. The VIP are the pieces that are needed to understand the research's own studied phenomenon, they are the key to this research itself. As a result, it is even intuitive and reflective of the philosophical paradigm. Interpretivism, explained by Rashid et al. (2019), is the paradigm of understanding the richness of social occurrences and values participants' own understanding as the piece of the development. However, one apparent disadvantage of the abductive approach that Awuzie and McDermott (2017) found is that there is a lack of concern for summarising the findings. It rather aligns itself with displaying its information richness with accompanying new ideas, than finding meaningful conclusions of it all.

3.2 Research Design

To get a research project out of the chosen research question (s) and aim is to have a specified research design. The method of the research design can either be qualitative, quantitative or mixed according to Saunders, Lewis and Thornhill (2023). Relevant aspects such as which data collection to use, what kind of time frame to set up and what techniques to use when analysing are all dependent on the chosen research design. For this research study, a qualitative research framework is most appropriate for exploring the meanings attributed to the participants. This is most appropriate according to the authors Saunders, Lewis and Thornhill (2023) because qualitative research involves collecting data through methods like interviews or observations. The data itself is non-numeric and can implement elements such as words, audio recordings, video clips and images. Ultimately, this is to analyse the data and use techniques like storytelling or pattern recognition to



finalise the findings. Essentially, this is a complementary method to the exploratory nature of the research questions leaning towards an open-ended approach.

Denny and Weckesser (2022) argue that qualitative research typically commences with the formulation of more comprehensive research question (s) which are subject to change throughout the research journey to refine and focus the study's aim or purpose. By opting for a qualitative design, this thesis can exercise flexibility and enable a more profound exploration of the challenges and opportunities present in the supermarket retail environment for individuals who are visually impaired. This also means that the research (s) questions can develop even further to embody the study's aim or purpose. The authors Nielsen et al. (2020) pointed out that in studies such as international business research, it has become more common with quantitative methods, which adopt longitudinal, large-scale data collection to study a phenomenon. They were critical towards the trajectory of how different phenomena are all studied in similar manners. It was conclusively argued that future research should be further encouraged to utilise a wider range of research designs instead of returning to the status quo.

3.3 Research Strategy

When a project has formed a designated research design by choosing to answer the research question (s) through a qualitative, quantitative or mixed method, it is advised to adopt a research strategy appropriate to that specific design (Saunders, Lewis & Thornhill, 2023). In this case, considering that this research project is opting for a qualitative framework, strategies such as case study, ethnography, action research, narrative inquiry and grounded theory are applicable. A case study strategy will inherently grant this project a good foundation for gaining knowledge and insight into the case of people who are visually impaired, which makes it the primary strategy. A case study is defined as researching a subject to gain an in-depth understanding of a particular phenomenon or topic where a specific event, organisation, group or person is the case for that research project according to Yin (2018).

As suggested by Yin (2018) it is of great importance for a case study to gain insight from settings or contexts in real life. Choosing this strategy over others gives us the means to gain a deeper insight into the phenomenon being studied. The author also states that case study research is effectively utilised by giving detailed empirical descriptions to develop new theoretical insight into the topic. When conducting case study research Bell, Bryman and Harley (2019) state that the researcher must be careful in what kind of sampling is done. The case (s) are required to be applicable to the research topic chosen for that project. The authors also refer to various discussions around the use of case study research and its reliability and validity. There exists a concern for the external validity of case



study research. This is something that this research project has taken into consideration and argued for under [3.9 Quality of Research](#).

Adopting a case study strategy together with an abductive approach comes with the possible advantage of enhanced theoretical strength argued by Rashid et al. (2019). According to the authors, an abductive approach accepts the pre-existing theory which can improve the case analysis more than if an inductive approach were chosen. Compared to deduction, abduction enables a more flexible research process which for this research project is heavily favoured when implementing a case study strategy.

3.4 Type of Data

The type of research strategy chosen for a research project also needs to consider what type of data is to be collected. For a study to be able to answer the stated research question it is required to gather some form of data, that being either primary or secondary data. Different fields require different methods to gather the best type of primary and secondary data suited to that particular research question (s) (Mazhar et al. 2021).

To characterise primary data Saunders, Lewis and Thornhill (2023) state that it is new, original data that is being presented for the first time in the current research project. The collection of primary data is through some form of direct communication or link between the researcher and the respondent (Mazhar et al. 2021). For this research project, primary data collection was utilised and consisted of semi-structured interviews within the qualitative research framework. The reason for choosing primary data was to get a deeper understanding and further discussion about how VIP experience supermarkets from their own point of view. Considering the scarcity of prior research on this specific subject, primary data held paramount significance in illuminating this understudied area more in-depth. Considering that different method(s) of primary data collection depends on various factors such as project resources, strengths and weaknesses of the researcher, type of sampling and much more (Mazhar et al. 2021). Hence, this research project will reflect on the decision of primary data by being delicate and significant throughout the paper.

According to Saunders, Lewis and Thornhill (2023), secondary data is defined as data that has previously been collected where the purpose is dissimilar to that of the current study. Collecting secondary data could be done through raw data that is mainly untouched, or compiled data which has already been summarised or distinctively selected. Consequently, the decision was made to exclusively rely on primary data. All data collection activities were conducted as an integral part of the research project. This approach ensured a thorough exploration of the topic, rooted in firsthand accounts and insights directly gathered from the target demographic.



3.5 Research Method

As a researcher, providing and collecting relevant data is important and the research method is a useful technique to ensure this collection (Saunders, Lewis & Thornhill, 2023). The purpose of the research should be taken into consideration when deciding the research method. Qualitative methods use focus groups, observations (Denny & Weckesser, 2022) or interviews that provide results that can potentially generate useful insights and understandings if the interviews have been successfully designed in a way (Rowley, 2012).

This thesis aims to understand and explore how the retail experience in Supermarkets can be enhanced through in-store design, resulting in interviews as the best-suited option (Rowley, 2012). Looking at it from a different angle, concerning the collection of data regarding how individuals perceive various senses, Yakhlef (2015) proposed adopting a methodological implications approach due to the intricate interplay between the body and the environment. It is advisable to utilise open-ended interviews to capture comprehensive experiences, aiming to uncover the nuanced, personal meanings individuals associate with their surroundings. This aligns with the qualitative methodology's focus on exploring the interpersonal dimension. Additionally, since the respondents have a visual impairment, interviews where the participants only have to use their voice and ears create greater accessibility to participate in the study.

3.5.1 Type of interview

Interviews can be divided into three categories: structured which are similar to questionnaires (Rowley, 2012); unstructured which addresses a few topics in great depth (Denny & Weckesser, 2022) and are only based on themes and requires a skilled interviewer as well as resulting in demanding result to compare (Rowley, 2012); semi-structured which are a combination of both structured and unstructured where the question order and the open-ended questions adapt to accommodate the interviewee (Rowley, 2012; Denny & Weckesser, 2022). Since the authors are novice researchers, semi-structured interviews are the recommended choice. Furthermore, to collect data best suited to answer the research questions semi-structured interviews were the preferred choice as well as enabled a wider range of possible discussions and answers.

3.5.2 Collection of data

An interview guide was established with the purpose of guiding the interview through important topics and providing relevant findings for the research. The authors were well prepared on the topic of the research when designing the questions in order to be relevant and beneficial for the study as well as the scientific methodology. Before the interviews, a pilot study with a visually impaired person was conducted in order to evaluate and adjust the interview guide with the purpose of receiving as valuable discussions and answers as possible. In addition, the pilot study provided perspectives and guidelines to facilitate good



conditions for well-founded and valuable empirical findings and analysis. Additionally, the participant provided guidance on the interviewer's conduct aimed at enhancing the ease and comfort of the participants throughout the interview session.

The interviews themselves were conducted between the 10th to the 24th of April, through Zoom, Microsoft Teams or telephone depending on which tool was preferred by the participants. Telephone interviews enable fewer distractions and participants tend to focus more on the voice and what is being said (Ward, Gott & Hoare, 2015), which can ease the participation and provide beneficial conditions for discussion. The research authors took turns conducting the interviews. During each interview two authors participated with different responsibilities, one led the interview and held the majority of the discussions while the other was responsible for the recording and took part in some of the discussion. The interviews were recorded by telephone during the telephone interviews and during Zoom and Teams interviews, the recording options via the utilities were used. Afterwards, each interview was transcribed.

3.6 Operationalization

During the establishment of the literature review a conceptual framework was identified which constituted a foundation for the creation of the earlier-mentioned interview guide. In order to provide guidance to the interviewers regarding the important topics for the research, the creation process was devised in conjunction with the research questions. The aim of using primary open-ended questions was to encourage participants to offer detailed and elaborated reasonings on various subjects. Following the format of semi-structured interviews, each individual interview was shaped partially by the informant's answers. Further, these answers were followed up by non-scripted sub-questions and avoided repetition by not asking questions which already had been answered by previously asked questions. The operationalization (Appendix C) is based on the thesis structure and concepts, however, the opportunity to provide new topics and suggestions is offered for the participants in order to discuss and explore how the experience and in-store design in supermarkets can be improved.

Additionally, to provide the best possible conditions for in-depth discussions and enabling well-elaborated reasoning the interviews were conducted in Swedish. By using this approach, possible difficulties with language barriers were avoided. Therefore, the interview guide exists in both an English version and a Swedish version (Appendix D & E). When choosing this approach, the possibility of errors was acknowledged which were considered carefully when the interviewee's responses were translated and transcribed. Moreover, both the pilot study and the supervisor were integral assistance when constructing the interview guide in the early stages in the project. This was to ensure both academic and VIP perspectives were covered in the interview guide methodically.



3.7 Sampling

Saunders, Lewis and Thornhill (2023) explain that sampling only exists because of the impossibility of obtaining every data source in one's work. This is why sample is used instead. To represent the margin. Considering the nature of this research is qualitative then it is reasonable that there is a notion of a non-probability purposive sampling. Bell, Bryman and Harley (2019) explain that the research questions should guide that decides what units are required to be sampled. In this case, the research question revolves around the visually impaired then the sampling by its foundation is purposive in that regard.

Intensity sampling is one sample method that this thesis will follow. It is about researching or learning from the particular rather than the regular and normal (Saunders, Lewis, & Thornhill, 2023). The aim for the sample is to enable the researcher to grasp the phenomenon occurring, while not resorting to the most extreme cases. Patton (2015) state that for intensity sampling to work, researchers are required to do some prior exploration to accurately judge what requirements are needed for the sampling. Understanding the rules of intensity sampling, this thesis picked intensity sample because of the nature of being visually impaired is a variation from the norm. The phenomenon of VIP's experience in supermarkets is the circumstance that this thesis wants to showcase.

One realistic opportunity that needs to be addressed is the fact that this research does take volunteers, which conditions the research to snowball sampling. According to Bell, Bryman, and Harley (2019), snowball sampling is the approach where the researcher makes an introductory contact with the relevant group, which opens up the opportunity to network with other members of that group through the initial contact. Essentially the identified member from the desired population helps to identify other members and so on (Saunders, Lewis & Thornhill, 2023). VIP is the target group, and it helps that members of that particular condition can recommend other members that experience similarly. However, a known issue is sources being biased because of chosen respondents being similar to the one recommended in for example opinion. Nevertheless, the population of VIP are inherently difficult to identify, and it helps that members of that group could guide the research to members with the same condition. See Appendix F to see the total number of interviews conducted, their anonymous names, the date, the formats and the length of the interviews.

3.8 Data Analysis

Bell, Bryman and Harley (2019) emphasise that one challenge that comes with conducting qualitative research is the extensive and intricate datasets that quickly build up when working with unstructured language. For qualitative research, the unstructured language is presented in the forms of documents, fieldnotes, interview transcripts or images. The authors continue that the richness of data can prove



difficult to analyse, making it a daunting task to find significant data for once research purpose.

The two most common methods and strategies for analysing qualitative data are grounded theory and thematic analysis (Bell, Bryman & Harley, 2019). To analyse the collected data and most accurately be able to answer the research questions a thematic analysis strategy was chosen. According to Braun and Clarke (2012), thematic analysis is both a flexible and systematic approach. It is a flexible strategy as it is not connected with any specific research philosophy, both subjectivist and objectivist positions could be applicable. What makes thematic analysis systematic is that it allows the researcher to get a logical and structured approach to analyse the collected data. Additionally, the time frame for the research project is better suited for a thematic analysis than that of a grounded theory approach. According to Bishop et al. (2020), a project adopting an abductive approach such as this one can effectively utilise thematic analysis. It enables the data to be explored and analysed in multiple ways rather than one preset approach. The thematic analysis was conducted following the six-phase approach presented by Braun and Clarke (2012). The approach consists of six stages that are gradually worked through in the analysis process, starting with Phase 1 of familiarising yourself with the data followed by Phase 2: Generating Initial Codes, Phase 3: Searching for Themes, Phase 4: Reviewing Potential Themes, Phase 5: Defining and Naming Themes and Phase 6: Producing the Report. A visual guide and representation of how this research project worked through the six phases is presented in Appendix G.

3.9 Quality of Research

Connected closely to the research design is the quality of research (Saunders, Lewis & Thornhill, 2023). When conducting the research, it becomes ever so important to understand and familiarise the importance of quality assessments, specifically for example the reliability and validity of the research itself. This is all to reduce the possibility of a circumstance where scrutiny may occur. This research thesis will discuss the quality of the research design with the help of Saunders, Lewis and Thornhill (2023), but also consider points discussed by Bell, Bryman and Harley (2019).

3.9.1 Reliability in Qualitative Research

Reliability is the consistency in the research process (Saunders, Lewis & Thornhill 2023; Bell, Bryman & Harley 2019). Researchers should be able to replicate and find the same kind of findings that occurred prior in another research through the same means. Only when the replicability of the research is ensured can it be assessed as reliable. Saunders, Lewis and Thornhill (2023) further explained that there are two subgroups of reliability: internal and external reliability. Internal reliability emphasises the coherence of the research project itself. In contrast, external reliability emphasises the consistency of environmental change affected by



the same procedures as data collection and analysis techniques. A few of the ways that increased internal reliability in this thesis was that the research authors were, in total, three people, which helps with the discussion of consistency in the whole thesis itself. This can also be reflected in the way that interviews were conducted. A rule of thumb was followed: to always try to have at least two members in each interview while taking turns as the one leading the interview and the one observing/assisting it. This held consistency throughout the whole interview phase while reducing researcher bias and error, on that end. The researchers also practise notetaking with the help of comments, writing down questions, thoughts and reminders alike. When it comes to external reliability, it helped that data saturation was the overall aim, and it can be reflected in the number of interviews and the variation they were conducted. This reduced participant error as well as helped with participant bias; the fact that the interviews were conducted either through different technological platforms gave the interviewees freedom to express their views without any definitive environmental factors that could affect the quality of the data.

3.9.2 Validity in Qualitative Research

Validity is about the integrity of the results from the research itself (Bell, Bryman & Harley 2019). However, Saunders, Lewis, and Thornhill (2023) do add that it also involves measurement accuracy. Validity can be divided into two terms: internal and external validity. Internal validity concerns the consistency of the phenomenon occurring and that it is related to the research findings. Techniques that were used to increase the integrity of the internal validity in this research outside of prior studies of the subject were both through discussion with the pilot study and supervisor. The supervisor helped double-check the integrity of the whole research and the integrity of the interview questions. Meanwhile, the pilot study assisted with the integrity of VIP actually being represented and connected to the subject matter. This decreased the testing and uncertainty about causal direction errors that might have occurred. Lastly, external validity concerns itself with the generalisability of the study being applicable in different contexts. It is essential to understand the extent of the generality of the research because of its possible replication. This research is about the VIP in supermarkets and their retail experience. Some practices applicable to VIP can be shared with BP, which may also be helpful for visually able people to an extent. The context of supermarkets is also very flexible in its generalisability. Practices that can improve the VIP retail experience can be shared with all other retailers depending on the practice itself and its applicability.

Bell, Bryman and Harley (2019) also expanded validity to ecological validity. Connecting the value that business research creates in everyday life. Considering that internal validity and external validity are already met, the VIP are the target group and what they experience is what the research seeks to unravel. These factors make this research ecologically valid. Furthermore, participant or member validation has been a close concept that this research has worked on (Saunders,



Lewis & Thornhill, 2023). Member validation is about reconfirming the quality and consistency of the data and its process through open discussion with the data source. This research has proposed to each interviewee that they can at any point take back their answers from the research and also take part of the research when it is done on their own accord. This is all to increase validation from the participants' side.

3.10 Ethical Considerations

Saunders, Lewis, and Thornhill (2023) emphasise the ethical conduct of research. Consequently, this thesis followed the ethical principles outlined by the authors in this research project (Saunders, Lewis, and Thornhill, 2023, pp. 256-257). First, the privacy of the interview participants and the insurance that they were informed about consent before their involvement was prioritised. Secondly, there is a guarantee of anonymity with the interviewee. Secure data storage is used to maintain confidentiality and anonymity. Thirdly, there is an emphasis on voluntary participation and clarification for the interviewees and their option to withdraw at any point during the project. Finally, all research findings are solely for this research purpose, and it will accurately report without any manipulation to alter the results. This also follows the concepts of secrecy, confidentiality, anonymity, and integrity mentioned by the scientific council (God forskningssed, 2017).

Moreover, it is crucial for the authors of this thesis to acknowledge the impact that this research can have and to accurately assess the potential benefits (Saunders, Lewis & Thornhill, 2023). This sets up the importance of the usability of research results and their connection to who and what they represent (God forskningssed, 2017). Before gathering empirical data through qualitative interviews, the authors engaged in thorough research to accurately comprehend the research domain and formulate significant research questions. Hence, comprehending the research and the potential results that it can have on VIP and supermarkets. Increasing the transformation of the research findings and analysis to actual practical use (God forskningssed, 2017). Furthermore, as the research prioritises the well-being of participants first, the methodology will be tailored to align with both appropriate scientific standards and the optimal experience for the participants (Saunders, Lewis & Thornhill, 2023).

3.11 Sustainable Considerations

In this research and associated activities, the aim was to minimise efforts that might result in a less sustainable report. The work defined sustainability according to the United Nations, which emphasises sustainability as “*meeting the needs of the present without compromising the ability of future generations to meet their own needs.*” (United Nations, 2024b). The study adopted the Triple Bottom Line (TBL) sustainable framework, considering environmental, economic, and social perspectives (Miller, 2023). From an environmental standpoint, the project



prioritised conducting interviews and gathering data through technological channels to minimise transportation, the need for resources and energy consumption. Additionally, refraining from printing or utilising physical products when digital alternatives were available. If physical copies were required, recyclable materials were used. Economically, expenditures were minimised by avoiding unnecessary purchases and striving to minimise transportation costs. Any project-related spending is subject to equal input from all involved parties to find the most suitable middle ground. The authors rejected attempts by companies or organisations to influence the report, ensuring the integrity of the research remained intact. From a social sustainability perspective as well as an economic sustainability perspective, this is a factor to consider. Socially, open communication is paramount, with all contributors listening to and respecting each other's perspectives. The work aims for equal inclusion and representation among all team members, with work distributed evenly to prevent the creation of a detrimental work environment.

The project aligned with several Sustainable Development Goals (United Nations Sustainable Development, 2024), including Goal 8 (Decent work and economic growth), Goal 10 (Reduced inequalities), Goal 11 (Sustainable cities and communities), and Goal 16 (Peace, justice, and strong institutions). It focused on improving accessibility for VIP in retail environments, specifically examining how supermarkets can enhance the VIP experience. Rooted in these Development Goals, the project aims to contribute to a more sustainable world by designing spaces with VIP in mind. Importantly, the report seeks to promote inclusivity rather than increase consumerism within society.

4. Empirical Findings & Analysis

In this chapter, the authors delineate a series of interviewees, subsequently elucidating the empirical findings derived from semi-structured interviews. These findings are then synthesised within each theme identified through thematic coding, aligning with the research questions. The data is simultaneously analysed during this chapter by connecting the empirical findings with the literature review. Similarities as well as differences between the literature and the empirical findings will be analysed and discussed. Furthermore, this chapter will discuss and analyse new perspectives to the field.

4.1 Informants

All participants in the study were guaranteed anonymity and will therefore not be named by their real names. In order to provide fluency for the reader as well as respect the interviewee's privacy, all participants were given pseudonyms of common Swedish generic names. These generic names are presented in an informant table (Appendix F).



4.2 Identified themes

As presented in the data analysis, thematic coding was used in order to identify themes from the collected empirical data. In the earlier stages, several different themes were identified, varying in quality and relevance to the project's research questions. After reviewing these findings, some themes were considered irrelevant while others could be combined, resulting in three coherent and distinctive themes: *The Physical Space of Supermarkets; Crucial Aids; Vip's Senses in Retail Experience*. In two of the three main themes, smaller sub-themes could be identified. The sub-themes have similar patterns to each other but answer the research questions in different ways.

4.3 The Physical Space of Supermarkets

During the interviews, the importance of how the physical space of supermarkets were designed in terms of layout, interior implementations and orientation emerged. The theme provides a new approach for in-store design, putting accessibility and facilitation for VIP in the centre of design, consisting of two sub-themes that outline the difficulties and opportunities the informants face when navigating and traversing the supermarket landscape.

4.3.1 Layout, Placement and Visuals

Johan¹: *"The first thing you think that there is maybe, that from store to store"*

This example is one out of many where the informants mention that the retail experience of supermarkets vary between supermarkets. One aspect that changes depending on which supermarket the informant visits is the physical space. The sub-theme *Layout, Placement and Visuals* showcases how the informants encounter these points of contact. When asked about their experience navigating the stores the majority mention shelves and displays posing as various obstacles in the store layout:

Erik²: *"So things that are difficult for example, it is when you have shelves that stick out at head height when there is nothing that sticks out at ground level, for example if you think, because if you walk with a white cane or you walk with a guide dog, then you primarily discover obstacles in, at a lower level because that's where you have the cane, as it were. You can't wave it around your head so that, That, it happens that you walk into shelves that stick out without there being anything at ground level... That's the point of the cane, that it hits them because you have to know what they are and you shouldn't go into them yourself, so then you have to make sure that the things you put on the floor don't somehow risk to fall over and crash too easily if you bump into them."*

¹ Johan, interview at 10th April, 2024

² Erik, interview at 15th April, 2024



Johanna³: *“but I think it's very bad if you place the cart or the cart or things like that in the middle of the road and so and or so yes, you know when they when the staff in the store would sort things and so, then they can, then it can happen in the middle of the road, it says in something like that. Then I won't detect it and hit it right away. Or when the shelves are arranged in such a way that goods almost stick out so that you can easily hit them with your shoulder and, I'm a little afraid of them.”*

The majority of informants state similar concerns to the examples. While not identical, similar patterns can be seen by sighted customers as presented by Mau, Schweizer and Oriet (2023) where retail stores should avoid placing products too high. For VIP this is not as straightforward as it is for sighted customers. The article argues that products that are too elevated come with the challenge of customers having trouble establishing a mental map of the environment. While this might be true for VIP in other aspects of the retail experience, products placed too high up pose a more dangerous issue where VIP not only run the risk of disorientation but potentially injuring themselves. This is more in line with Yu, Tullio-Pow and Akhtar (2015) and their view on tripping hazards for VIP which the informants also describe:

Helena⁴: *“But the shelves feel strange, sometimes there is a pillar in the middle of the aisle when you have to walk instead. It feels like it's not really working. That they hadn't really thought. I don't know how they thought... sometimes I think there are too many such baskets on the floor. Maybe these offers are a bit like that, but sometimes I think it's a bit difficult when they, you're not prepared for it to get in the way when you want to go”*

It is crucial for supermarkets to prioritise layout that emphasises clarity and simplicity to make it more accessible for VIP as argued by Yu, Tullio-Pow and Akhtar (2015). In the interviews, informants describe how the aisle can be designed to improve the layout of supermarkets.

Erik⁵: *“Decent like straight corridors that it's not that much, so that you think that you build a little on grid patterns to some extent of it. That you don't have too much, like corridors that go in different directions... so that you can orient yourself a little easier... if you use some kind of decent grid pattern in straight corridors in one direction or the other, it will be easier.”*

Fredrik⁶: *“It can be very very narrow. Then people come with these big shopping carts and stuff like that. And, yes, it can get a bit crowded so it can be a problem, but then I try to bring white cane and that usually helps so they move on.”*

³ Johanna, interview at 15th April, 2024

⁴ Helena, interview at 17th April, 2024

⁵ Erik (n 3).

⁶ Fredrik, interview at 17th April, 2024



It is not only Yu, Tullio-Pow and Akhtar (2015) that mention similar aspects as those stated by the informants. Ebster and Garaus (2015) describe how narrow aisles should be avoided as customers feel discomfort encroaching on others personal space. The article focuses on sighted customers and Customer Traffic but evidently, by the gathered data a comparable analysis could be done for customers who are visually impaired as well. In many of the interviews the interviewees talked about feeling discomfort when visiting supermarkets. The reason for that discomfort came in the form of potentially disturbing other customers and their personal space.

Johan⁷: *“If you think about that way of narrow (aisles) and a lot, if there would be a lot of customers in the store, then maybe you would avoid those moments or it would be, make the experience worse than it is. And narrow, yes, corridors.”*

Fredrik⁸: *Sometimes I feel like I'm disturbing people when I stand at a shelf for 5 minutes, just standing looking for an item, so I try to hurry because there can be a lot of people in the stores sometimes and so and also when they're walking around in the aisles.*

The research's literature review presents numerous publications discussing the topic of retail stores physical space and previous research done on customers who are visually impaired. However, when analysing the literature with the research's empirical findings there is a lot to be desired. A great number of informants expressed dissatisfaction in trying to navigate the supermarket. Corridors and walkways being littered with obstructing objects and aisles designed too narrow “you have to walk not normally like this. You almost have to walk with your shoulder forward” (Johanna). Supermarkets that utilise in-store design with similar layout does not help VIP who already find it exhausting to go to the supermarket: “in the end you get tired so you just leave the grocery store and stay at home instead” (Anders). The data gathered in the research tell a different story than that of Balconi, Acconito and Angioletti's (2022). When expressing their own experience of supermarkets VIP do in fact find it more difficult to search for specific items. The stated literature proposes that VIPs' abilities to search for items do not differ in a major way from sighted people. What the article might be missing is that no supermarket is designed the same. If a visually impaired customer is unable to traverse the supermarket and search for a specific item due to the in-store design, the ability cannot be the same as for a sighted customer.

The physical space of supermarkets also includes placement of aisles, shelves and products as discussed in the literature review. The experience for VIP is largely dependent on how supermarkets utilise shelf management in in-store design. Rearranging the product positioning and layout of the store posed as one of most prominent challenges for VIP in supermarkets:

⁷ Johan (n 1).

⁸ Fredrik (n 6).



Anna⁹: *“Sometimes they have a tendency to move things around so that once you have learned where different products are, they can then change it a bit and then you have to learn again and then it can take quite a long time and to go and shop because you go and search and so on.”*

Olof¹⁰: *“The downside is when they change it. I find that very disturbing and I think most people find it very disturbing. Then you sort of learned the grocery store because you often have a grocery store that is like your favourite store or what to say and. That's how you've learned where the goods are. And then all of a sudden they just come and change”*

Erik¹¹: *“Supermarkets have a tendency to rearrange layout at regular intervals, for example, to make people walk around a little more and find more things they're hungry for and it's not really adapted to us who don't see”*

Sara¹²: *“And it's not worth it for me to try to find items in a store where I don't know the structure”*

Close to all of the informants talked about how permanent placings of products and a consistent layout improves their retail experience in supermarkets. The gathered data complement the findings of Kostyra et al. (2017) where avoiding changes to product placement facilitates the shopping experience for VIP. Supermarkets who stick to a specific layout and have permanent shelves become more accessible than supermarkets that do. This could be seen as a risk if supermarkets follow the findings of Drèze, Hoch and Purk (1994), Bianchi et al. (2021) and Valenzuela, Raghubir, and Mitakakis (2013) who argue there exists a potential financial gain in temporary shelves and displays. Supermarkets that implement shelf management like the mentioned articles may improve the experience for sighted customers and gain financially. The question then could be asked if the financial gain and improvement of retail experience for sighted customers is greater than the damaged experience for visually impaired customers. The articles also state the effectiveness of permanent displays and shelves, which for VIP is beneficial. Supermarkets could find strategies that allow them to gain the benefits of temporary and changing layouts without the cost of a worse experience for customers who are visually impaired.

The physical space of supermarkets does not end at floor level but also includes the store's lighting. The gathered data on how the lightning in supermarkets is not as straightforward as earlier discussed points. Some of the informants stated that the lightning plays an important role and that the experience differs from various supermarkets:

⁹ Anna, interview at 11th April, 2024

¹⁰ Olof, interview at 17th April, 2024

¹¹ Erik (n 2).

¹² Sara, interview at 15th April, 2024



Louise¹³: *"I'm having a hard time. I have to have a lot of light into a room if it's a little semi-dark, then it just gets dark. Then I hardly see anything... I need very bright if I'm going to look at something with a magnifying glass"*

A few of the informants even got affected by what type of lightning the supermarket uses:

Sara¹⁴: *"There must be good light. Very bright and so for that, it should preferably be led, if you have a led plate then no flickering fluorescent tubes and the like....Now you have come so far because the lighting is inside the freezers. There is nothing that dazzles, it must not dazzle the light. It's allowed, it should be indirect light... the light should be there but you shouldn't see it. You shouldn't see the fixture if you say that."*

On the other hand, the collected data showed that some VIP do not get affected by the lightning in supermarkets:

Magnus¹⁵: *So it (lightning) doesn't really affect that much, because I, like, I don't use it as much as I notice if the lights go off or if they're already off or if they're on. But it doesn't affect me that much, like orientation or during the stay itself"*

The effects of in store lightning varied between the informants but when analysing the data it showed that the majority of informants express that the lightning did affect their retail experience and is an important aspect. These findings stand in contradiction to those findings of Quartier, Christiaans, and Van Cleempoel (2008) who argue that consumers do not get affected by lightning and are unaware of the differences. Their article focuses on sighted consumers but for visually impaired consumers the collected data shows more varied findings with a tendency to be that lightning is something VIP do notice. The findings of this research are more consistent with Yu, Tullio-Pow and Akhtar (2015) and matches their understanding of how in-store designs lightning affect VIP. Like the article, the collected data shows that well-lit, consistent lighting and clear contrasts from various surfaces enhance the retail experience. One factor that needs to be considered is that the informants did not all experience lightning the same. This could prove to be difficult for supermarkets as it may be hard to find universal lighting that works in every store and for all VIP.

The contrast in supermarkets was not only found to be a factor surrounding the lighting for VIP but also on signages. A large part of the informants talked about how contrast on signs in supermarkets is an important factor when navigating the store and searching for products. This is also consistent with the findings of Kostyra et al. (2017). What makes it difficult when it comes to signs and their

¹³ Louise, interview at 16th April, 2024

¹⁴ Sara (n 12).

¹⁵ Magnus, interview at 15th April, 2024



contrast is that the informants stated different needs from each other. Everyone's experience was their own and found different characteristics challenging. Some informants experienced it difficult when supermarkets utilised varying colours on signs whilst others found it difficult when the signs consisted of low contrasts:

Stefan¹⁶: *“Yes, the contrasts are a text of very great comp, importance... make sure they have text with good contrast because it's not just me who is visually impaired... And it's almost impossible to read it because it's gray on white background.”*

Gustav¹⁷: *“It's colours because I can't handle and such. Often they have big red signs with something and I absolutely cannot read that. ...It has to be black and white so that I can read anything at all, and it's often advertising stuff. So there's a big sign saying that this costs so and so, but I can't read that”*

Sara¹⁸: *“Contrasts are very important. It is very important, not too many mixed colours.”*

Viewing the data collected from the all informants show a resemblance to the articles of Kostyra et al. (2017) and Yu, Tullio-Pow and Akhatar (2015). Whether the contrast needs to be colourful or black and white, the text and background requires to have clear distinction between each other to be useful for VIP. Furthermore, every informant that could implement signs in their retail experience noted that large fonts on letters is imperative for a smoother experience, regardless of colour. *“The contrast is good there, but the text is small, so it's hard to read”* (Anna). However, signs that have a clear distinction and large letters can still be of no use for VIP if their placement is not thought through with accessibility in mind:

Fredrik¹⁹: *“I think it is important that it is signposted well. That, there are clear signs that show what is on the shelves... it's quite a good height and good text, clear text as well... So like, preferably like, dark background and so wide text... And that it is a very bad sign that the signs are also hanging very high up. It's hard to see them”*

Sara²⁰: *“They usually sit so high that I don't enjoy them”*

The collected data stand in good relation to the findings of Baker, Stephens and Hill (2002) that the placement of signs is just as important as the contrast on them. By analysing the gathered data with previous literature, supermarkets should avoid placing signs high up and above the middle of aisles as this excludes VIP. The retail experience worsens for VIP when time and energy need to be spent on localising specific aisles or shelves with no assistance or cues from the physical

¹⁶ Stefan, interview at 12th April, 2024

¹⁷ Gustav, interview at 24th April, 2024

¹⁸ Sara (n 12).

¹⁹ Fredrik (n 6).

²⁰ Sara (n 12).



space you are in. A portion of the informants also expressed how supermarkets have started implementing digital signs in their stores, implicating findings with similarities to Yu, Tullio-Pow and Akhtar (2015). Digital signs may not be a solution for VIP at this point in time as these signs suffer the same problem as traditional ones with bad contrast and unreadable text:

Anna²¹: *“Some even have digital signs in some shops. They are very difficult to read... It's poor contrast.”*

Pure visual signs could therefore prove challenging for many VIP and some informants expressed that no matter how well a store implements visual signs, they are still unable to positively affect the retail experience. One aspect that many brought up during their interviews was the potential implementation of tactile signs:

Fredrik²²: *“that's exactly the thing about tactile, then. I think they write something like this on the signs and things like that maybe on each shelf then write something like this there is this there is this here. Either braille or such tactile letters that you feel the letters.”*

Johan²³: *“You might have been able to if you had signs, i.e. braille signs at every aisle or so. For example where you name a bit, yes But this corridor can contain lalala lalala. If you hadn't been able to have it on every single item that in this, or on this shelf you can find these items, it could have actually made it easier too”*

Johanna²⁴: *“somehow do, put braille on the shelves so that you roughly understand what is on this shelf. Either with buttons or with those braille characters that you can recognize roughly that way”*

Tactile signs would improve their experience where navigation and search for specific products becomes less of a daunting task. This would allow supermarkets to become more accessible in more than one way and implement in-store design better suited for a wider range of customers. All the points brought up in this first sub-theme could of course be analysed and potentially integrated together as the physical space of a supermarket is not divided into certain groups but rather one universal part of the retail experience.

4.3.2 Tactile Implementation

In the second sub-theme *Tactile Implementation*, the use of tactile to eliminate barriers for VIP is identified and analysed. As discussed in the first sub-theme, obstacles have a problematic impact on VIPs' retail experience in supermarkets. These obstacles could be reduced by implementing different tactile techniques,

²¹ Anna (n 9).

²² Fredrik (n 6).

²³ Johan (n 1).

²⁴ Johanna (n 3).



such as tactile walkways. Many of the informants use a white cane for navigation and identification of possible obstacles to avoid. By adding tactile walkways, this process could be facilitated:

Johan²⁵: *"you might want to avoid getting into things, maybe it would have been easier if you had a line that you could follow with the cane and that it turns where the corridor, how do you say, turns"*

The majority of the informants express similar thoughts, findings which are supported by the literature (Yu, Tullio-Pow & Akhtar, 2015; Celik & Yakut, 2021) and strengthens the suggestion for tactile walkways as a beneficial implementation for retailer to enhance VIPs' emotional experience (Balconi, Acconito & Angioletti, 2022). Additionally, the authors argue that such sensory cues enhance the recognition process and could also be seen as a marketing strategy. Therefore, the implementation of tactile walkways not only increases accessibility in terms of navigation and product recognition but also provides the retailer with a strategy for attracting new customers. In addition, tactile walkways can be argued as more beneficial than carpets as directional cues, since tactile walkways do not include the same risk of falling as carpet edges does (Yu, Tullio-Pow & Akhtar, 2015). Yet, one informant argued that tactile walkways could be problematic while walking:

Gustav²⁶: *"it would be uneven in the aisles in such cases. I don't think they would work at all."*

Another identified disadvantage connected to tactile walkways is differences of opinion regarding where the walkways should be placed:

Sofia²⁷: *But, I don't think you can demand that the store make lots of walkways on the floor that go here or there. Some would like the walkway to the milk and others would like it to the flour.*

Erik highlights another perspective on the implementation of tactile walkways, emphasising the importance of a well thought through implementation:

Erik²⁸: *"it would certainly be possible to make some kind of tactile walkway in, for example, the ground that you can follow with the cane... of course, you can't have it in every aisle everywhere, like in a pattern. But maybe you can have something in the larger aisles, or maybe that you can mark where the cash registers are and where the entrances and exits are"*

Expanding this perspective further, a non-successful implementation would complicate instead of facilitate the orientation and navigation:

²⁵ Johan (n 1).

²⁶ Gustav (n 17).

²⁷ Sofia, interview at 11th April, 2024

²⁸ Erik (n 2).



Johanna²⁹: *“I actually really appreciate them, if they are placed correctly of course because it can happen that there are walkway, which lead to the shelf and then under the shelf... but if they are quite plain, easy to use.. and placed in a good way so it can be quite, it can actually be good”*

These four statements exemplify a few challenges which could contradict the efficiency and ease provided by tactile walkways. This problematization with tactile factors is also explained by Guo and Bigham (2018), arguing that simply adding tactile features is not enough to enhance accessibility. In addition, the contradiction would provide even greater challenges than the already existing ones, for example with product orientation (Kostyra et al. 2017). Contrary, tactile walkways would enable a certain increased flow of information when navigating, which is called for by VIP (Plikynas et al. 2020). Therefore, creating a plan of how implementation could be done in the most effective and prosperous way should be considered as well as research on how tactile walkways need to be designed to facilitate in the best way. Thus, the knowledge of the impact and thoughts from VIP are crucial.

As an alternative for tactile walkways and carpets, different kinds of floor coverings and tactile marks on already existing visual guidelines on the floor are suggested. Different floor covering could be used as a tool for the product recognition process:

Sofia³⁰: *“It would probably make it easier because you know approximately that yes, it is on the inner floor covering, that is the rough one. There are always, for example, baking ingredients or preserves”*

This tool would avoid the risk connected to carpets (Yu, Tullio & Akhtar, 2015) and “uneven aisles” (Gustav) but, on the other hand, would not provide a safe path to follow without obstacles or guidance to a specific product. Therefore, arguing for the positive impact of implementing different floor covering along with tactile walkways is well substantiated, however, additional in-store design elements such as accessible signs would also be preferable to facilitate the retail experience in supermarkets for VIP. Consequently, developing the already existing visual guidelines by adding tactile marks would increase the accessibility for VIP, facilitating the navigation:

Magnus³¹: *“There you could perhaps have a little more clear tactile markings that you can feel with the cane and follow like, if I go there then I'll get to the checkout or if I go there I'll get to, like, this specific department”*

²⁹ Johanna (n 3).

³⁰ Sofia (n 27).

³¹ Magnus (n 15).



During the interviews, it became clear that tactile markings would be preferable in several situations, not only on floors. These markings could facilitate both navigation and product recognition, as well as facilitate the use of technological aids such as mobile apps, findings which are not identical but describe similar advantages as Guo and Bigham (2018). Several of the informants compared food products with pharmaceutical products, where the use of braille enabled independence as well as reduced time while searching a specific product:

Johanna³²: *“the medicines usually have braille on them and that's actually great... You know the kind of non-prescription medicines that you can choose to check yourself, then it's a little better with the pharmacy that you can go and feel the braille and read the name of the item itself.”*

Johan³³: *“In any case, it would have helped. Sure braille isn't the whole answer but... if I went to a shelf and felt that here's a product, I could read what it is... So it would have gone easier, faster than if I maybe had, for example, instead of taking a picture of every single thing that I think is this, I could have ruled out. But here it says milk, it wasn't milk I wanted and I don't even have to take a picture with mine, with my help from the app.”*

In contrast to these statements, some of the empirical data highlights the lack of knowledge as it comes to reading braille, therefore, the view on the impact of braille has a distinct contrast. Opinions on braille as helpful differ from *“not at all, because I can't read braille”* (Anna) to explaining that VIP who were not born blind *“are not terribly bad at braille but we are not that super fast or skilled either”* (Sofia) as well as it could be helpful to find the right product but *“I still think you need to know roughly where to look, so to speak because you can't go and try to read every package in the entire store”* (Erik). Comparing these findings with Kostyra et al.'s (2017) and Slade and Edwards (2016) studies, it becomes clear that braille is not the most suitable change for increasing the accessibility for all VIP in supermarkets. However, implementing braille would facilitate and develop the product recognition process for those who can read it, saving both time and energy for the VIP while grocery shopping. In addition, braille would enable VIP to exercise their own independence (Guo & Bigham, 2018; Yu, Tullio-Pow & Akhtar, 2015). Since existing technological aids are better adapted for people who lost their sight over time (Boldu et al. 2020), braille would provide a similar alternative for those who are born blind, increasing the accessibility.

Instead of implementing braille for all product information, tactile marks could be argued as a more suitable strategy, findings which matches Kostyra et al. 's (2017) braille suggestions. For example, having the first letters of the product in braille would enable *“to rule out or know what it is.”* (Johan), facilitating the process. For people who are blind, braille enables them to read on the products which *“is*

³² Johanna (n 3).

³³ Johan (n 1).



absolutely decisive” (Sara). Further, one informant suggested tactile maps at the service desk with information on where to find different product categories, a strategy which not only would facilitate the orientation but also enhance the emotional customer experience (Balconi, Acconito & Angioletti, 2022). Similar to the implementation of braille on food products, the empirical findings indicates that when it comes to self-checkouts, braille could develop the accessibility, counteracting these interfaces inaccessibility (Guo & Bigham, 2018; Yu, Tullio-Pow & Akhtar, 2015).

Johanna³⁴: *“the self-service when you scan the food yourself and weigh it. There I think it might be good to have braille at the buttons if it is simply built and not that difficult, so that you know that there is the machinery that you can press and so on.*

In conclusion, braille in combination with other facilitating strategies will both ease the recognition of products and create a higher level of independence for VIP, resulting in a less time and energy demanding retail experience. The theme of *The Physical Space of Supermarkets* is a vital part in in-store design’s affect on the retail experience regardless of if the customer has a disability or not. As argued by Dodds and Palakshappa (2022), accessibility needs to be in the centre of design to be an inclusive environment and something these two sub-themes have showcased.

4.4 Crucial Aids

During VIP’s retail experience, aids play a crucial role for both accessibility and facilitation of the retail environment. The theme *Crucial Aids* provides insights on two crucial categories of aids and will discuss the importance of enabling these categories to interact with each other. These categories will be presented in the two following sub-themes: *Physical Aids* and *Personnel Aids*.

4.4.1 Physical Aids

As discussed in the literature review, the research and development of facilitating aids aimed towards VIP is large. In this first sub-theme *Physical Aids*, the existing as well as possible future physical aids will be presented and analysed. The empirical findings show that the most usable aid is the telephone and its apps, such as the app Be My Eyes. These technical aids can be designed in different ways and provide assistance through multiple methods:

Johanna³⁵: *“There are tons of apps that can photograph the product, read and say, what product it is by the name and so on. Or you can put on the app be my eyes, where you use video so that another sighted person can say ah, this is in front of you”*

³⁴ Johanna (n 3).

³⁵ Johanna ibid



Erik³⁶: *“There is an app called Be My Eye or Be My AI. I don't know if you've heard of it, but it is actually an AI-based image recognition so that if you take a picture of a milk package, it reads out what it is. There are also different apps or where you can, where you can get assistance from a sighted person with whom you connect. Which again uses the camera as well.”*

Anna³⁷: *“if it says on signs and such, then I usually use my phone. Take a picture and then enlarge to see what is written”*

One informant described a useful application designed for scanning the product's bar code and QR code and then providing the user with a short description of the product. Previous studies have predicted smartphones as the most probable future solution (Plikynas et al. 2020; Celik & Yakut, 2021), a development which is somewhat confirmed by the empirical findings in this study as. Additionally, smartphones achieve needs for VIP such as discreteness and readily available when needed (Boldu et al. 2020). However, the majority describe similar difficulties as Plikynas et al. (2020) with these aids, expressing a need for something more suitable. A conclusion can be drawn, that the predominant use of telephones and application depends on the lack of better and more flexible alternatives, not that the existing aids are good in themselves. In addition, VIP are required to *“try to find solutions yourself, in the grocery store”* (Anders) which leads to a feeling of exclusion from society, a feeling which could be counteracted by well adapted service quality, store accessibility and store ambience (Celik & Yakut, 2021). When using the existing aids, discomfort and irritation are common:

Gustav³⁸: *“I have received a camera from Syncentralen, but I can't handle it, because it is far too tiresome and pressing buttons here and there, I can't handle it.”*

Johan³⁹: *“the app that is currently the most popular takes a little time, but that's where you get the most description because it uses AI functions... get very good descriptions. But that being said, the process itself can take up to 20-30 seconds.”*

Linnea⁴⁰: *“you don't take out your phone just to show the staff that I can indeed scan in that this item... It takes too long”*

Even though these aids are designed to facilitate independence, grocery shopping for a VIP on its own takes up to three times longer than a sighted customer, according to one of the informants. These findings provide a new perspective on technological aids, implying that despite the reported usefulness of aids reading labels (Kostyra et al. 2017) the development does not fulfil VIP's needs. The time and energy spent weigh so heavily that, in the context of grocery shopping, VIP

³⁶ Erik (n 2).

³⁷ Anna (n 9).

³⁸ Gustav (n 17).

³⁹ Johan (n 1).

⁴⁰ Linnea, interview at 12th April, 2024



neglects their own independence mainly because of the aids lack of accessibility. According to Plikynas et al. (2020), this is a major factor for slow development in the area of research, which is contradictory since the phenomena highlight the importance for development. Further, asking for help is more accessible for VIP than using the existing aids (Boldu et al. 2020), reinforcing the informants demand for a development in this area.

As earlier discussed, braille maps and tactile guidance would be a functional alternative when facilitating for VIP during the retail experience in supermarkets, however, creating a database linked to a map has been suggested by several informants. Similar to the structure at department stores with a take self-stock, each shelf should be provided with a number. When searching for a specific product in the database, a description of which shelf the product can be located together with guidance would emerge. Additionally, the database would fulfil Boldu et al. 's (2020) crucial factors of technological aids for VIP, indicating the aids as highly relevant and effective. Plikynas et al. (2020) describes VIP need for information if environmental elements within a 20 metres range, a need for which the database could satisfy. Creating a database with the entire store structure programmed in, the GPS function would be able to notify if one was approaching any obstacles or walls, providing an orientation option which does not exist today. Even though the risk of colliding with other customers still exists, the risk of colliding with the store's in-store elements is eliminated. At last, similar to the implementation of tactile walkways, the importance of up-to-date structure is crucial.

Johan⁴¹: *“some form of indoor GPS or navigation if you had, for example, entered that I want to go to the bread shelf and then received some form of guidance when entering the store with the help of a GPS voice”*

Sara⁴²: *“that you have coordinates...where it says shelf, number. You can tap on an item if you have the app, so it says coordinated aisle c, left, row, shelf 4 for example and stuff like that. Just like it does in many speciality stores like this. Then it would be easier”*

From a more general customer perspective, implementing a database with so called coordinates would not only increase the accessibility for VIP but also provide a facilitating service for all customers when shopping grocery products. The statements above are supported by Celik and Yakut (2021), who suggest the creation of a multisensory environment, facilitating navigation through audible technologies and stores own application. From the perspective of VIP, audible cues are mainly seen as possible facilitators for product recognition. First and foremost, speech support on the handheld scanner *“So it reads out what you scanned”* (Sara) would inform directly the product and price, saving time as well as providing VIP the ability to be sure of buying the right product. This suggestion is supported by

⁴¹ Johan (n 1).

⁴² Sara (n 12).



Yu, Tullio-Pow and Akhtar (2015), arguing for the possible facilitation of handheld scanners functioning as an identification tool by either audio cues or read-aloud functionalities. However, the sound level in the current environment impacts the efficiency of audible functions (Boldu et al. 2020), which reduces the possibility of relying on such an aid. Using headphones could increase the possibility of using auditory cues but since most of the informants named sound as a crucial factor to decipher the surroundings, headphones would most likely reduce perception and thus cause greater disadvantages than advantages. Further implications of audible cues could be added at, for example, self-checkouts:

Anders⁴³: *“speech synthesis in the self-checkout...so that you can go in yourself and pay yourself and so you can scan products yourself.”*

4.4.2 Personnel Aids

The second sub-theme *Personnel Aids* will provide a discussion on the in-store personnels importance. Regardless of existing and future aids and for VIP, the need for and importance of personnel in supermarkets is of great importance. Kostyra et al. (2017) reports sales staff as a major enabling factor for grocery shopping while Yu, Tullio-Pow and Akhtar (2015) defines staff as a crucial factor for VIP’s retail experience, statements which the empirical findings of this study support. Even though in-store improvements and development of physical aids, combining these with personnel would be the most effective strategy for facilitating the retail experience for VIP:

Sofia⁴⁴: *“I think it's hard, but a combination of physical improvements and, so to speak, additional personal help in business could actually make it easier.”*

Helena⁴⁵: *“It is probably difficult to make it work technically without staff”*

The statements above strengthen Guo and Bigham’s (2018) statement regarding an integrated system between physical and personal aids. Other informants have expressed similar opinions and therefore provide an interest for comparing the different types of aids against each other. Both Boldu et al. (2020), Plikynas et al. (2020) and Guo and Bigham (2018) highlight the dependence on personal assistance in the context of accessibility for VIP. The findings of this study imply the same opinions, saying that the factor with the largest positive effect is *“if you meet or make good contact with the staff.”* (Johan), however, locating the personnel is often problematic, especially in larger supermarkets. This factor affects the choice of supermarket, by indirectly forcing VIP to select a smaller store where the possibility of help is better:

⁴³ Anders, interview at 11th April, 2024

⁴⁴ Sofia (n 27).

⁴⁵ Helena (n 4).



Erik⁴⁶: *“so that generally speaking, smaller businesses are easier....to be able to get help as well. Because the staff may not have quite as many things they are trying to do at the same time”*

These findings have similarities with Kostyra et al. (2017) result on supermarkets deficiencies in force providing personnel assistance. Beyond this, the informants describe a feeling of exclusion since smaller stores have a limited assortment and often higher prices and therefore, narrowing down the possibility to choose distinctly. The marketplace role as a opportunity to exercise independence for VIP (Baker, Stephens & Hill, 2002) decreases as the availability for assistance reduces, exemplified by:

Sofia⁴⁷: *“Above all, I would actually call for some form of more personal assistance to be provided in somewhat larger department stores...where the cheap prices are, where the large selection is found, and these are the stores we avoid”*

Contrary to previous research stating that asking for assistance is uncomfortable (Kostyra et al. 2017), these findings indicate the ability to ask for help as a crucial factor and the deficiency of additional aids increases the need further. By using auditory cues, VIP can identify a person standing close to them, without knowing if they have found personnel or not. For VIP with some sight left, the locating problem still exists since *“some people are colour blind and don't see like there is staff, there is a person.”* (Anders). Providing a direct solution for this problem seems difficult since each situation is individual which is consistent with Doods and Palakshappa (2022) findings, however, three beneficial factors for managing personnel aids have been identified: attention from personnel; treatment and education for personnel; additional shopping service. By encouraging the personnel to be attentive towards customers, VIP in need of help would be noticed quickly without spending too much time and energy on locating help.

Erik⁴⁸: *“It's really about the fact that there might be someone inside, like, who sees when you enter the store and can ask, do you need help? So that, it's hard when you can't see or can't see well to actually find someone to ask for help, so it is. It is based a little on the fact that they have some overview of who comes in.”*

The second factor highlights how the impact of treatment towards visually impaired customers could be enhanced by education. In addition to the general characteristics such as service minded and friendly approach, VIP expresses a desire for personnel with knowledge on how to facilitate and help for a person with a visual impairment. For example, guiding the visually impaired customer patiently to the desired products, avoid pointing towards the products and walking in front of the customer. These findings have similar patterns to both Baker, Stephens and Hill (2002), Nguyen et al. (2020) and Yu, Tullio-Pow and Akhtar (2015), however, Yu,

⁴⁶ Erik (n 2).

⁴⁷ Sofia (n 27).

⁴⁸ Erik (n 2).



Tullio-Pow and Akhtar (2015) also argues for knowledge of the stores' products as equally important. Increasing the knowledge could easily be managed through more and developed training for the personnel, which according to Celik and Yakut (2021) indicates a differentiation business strategy.

Johanna⁴⁹: *"You can improve the treatment of the staff and so... the staff should perhaps have some way, some, some knowledge that if there is a situation where someone is blind, they should come and help and check and ask if it is needed any help. Because I've come across once...I just stood like this, looking around and trying nothing and like this I went on for like almost an hour without getting any help"*

Anders⁵⁰: *"when they walk, they walk in front of you and you have to try like, they are going very fast so you will try to catch up, I can't see the person. But just hear how his, you like hear his or her footsteps to following after, but some staff were just "Come with me" and so they hold your hand and goes with you like wherever, wherever you want... It is noticeable that some staff have a lack of knowledge about people with visual impairments...They don't know what a visual impairment means, they don't know what a blind person means and they don't know...Understand what visual impairment is and help them in the best possible way."*

The third factor appears in the form of a suggestion of a service in supermarkets, to offer a shopping service for VIP. This service would enable a time- and energy saving option as well as the possibility to visit supermarkets without a companion. Different suggestions of how this service preferably would be designed exists, but several informants bring up the importance of adaptation for both the retailer and visually impaired customers. From the perspective of the retailer, factors such as time of day and different days of the week are meaningful while waiting time, type of procedure and information are important factors for VIP. Providing a bookable service with suitable time slots and personnel meeting up and guiding the customer through the store, both facilitating the product orientation and informing on potential extra prices. Additionally, this would enable VIP to try different brands of the same product, an opportunity that today barely exists due to the limited access of information resulting in forced brand loyalty (Kostyra et al. 2017).

Anna⁵¹: *"what the store could do is that they offer the service, a service that you can get help from some staff to shop....Yes that you can call and say that I would like help when I arrive and can you help me to buy what I need."*

Magnus⁵²: *"a member of staff...someone stands outside the store and waits because I have called in advance and said that if I arrive at this time."*

⁴⁹ Johanna (n 3).

⁵⁰ Anders (n 43).

⁵¹ Anna (n 9).

⁵² Magnus (n 15).



Even though the service would be beneficial for VIP in every store, the service vitality increases the larger the supermarket is since the *“staff in a smaller store are more helpful, or they probably have more time”* (Anna). There is a lack of earlier research on the mentioned topic of additional personal service and something that could greatly benefit from further development.

Lastly, some informants request the possibility to get assistance while shopping spontaneously, by having a service close to the entrance where *“You can press a bell”* (Louise) and personnel will show up and helpfully guide the customer through the store. However, the implementation of such services could be problematic, depending on the varying available personnel and customer density in the store. The discussion on personnel and their impact on VIP's retail experience have identified several factors of both positive and negative nature, expanding Taylor et al. 's (2019) theory on retail customer communication by identifying new facilitators and barriers. However, none of the informants have expressed experiences of unpleasant treatment, only lack of knowledge, findings which pose a distinct contrast to Taylor et al. 's (2019) factors for barriers. This implies that retailers' service policies for positive attitude towards customers is well functioning but needs development of treatment of customers with disabilities, which would also increase the overall inclusion (Dodds & Palakshappa, 2022).

4.5 VIP's Senses in Retail Experience

Erik⁵³: *“It's always a question of interplay between, in between the sensory impressions you get, regardless of whether you have sight or not, it's not like you only use one of the senses or it's very rare that you do, I think.”*

The statement is an example of how the senses interplay with the retail experience for VIP. During the last identified theme *VIP's Sense in Retail Experience*, VIP's use of different senses in supermarkets will be explored and discussed. According to Yakhlef (2015) the environment, the individual's senses, and their inputs are connected to one's perception. Each sense should be considered to understand the whole puzzle of human perception and how the different senses are congruent and incongruent with each other. The following statements provide an initial understanding of the intricacy of multifaceted senses in supermarkets for VIP:

Erik⁵⁴: *It's always a form of interaction so that it's difficult to kind of distinguish certain things that way... And then for me it's about verifying in different ways that we're in the right place and it can be. It could be the hearing. It can be like that, the perception of space you have if it then depends on hearing or on...Because it's both, it's both hearing and it's touch and it's maybe sight in the sense that you have*

⁵³ Erik (n 2).

⁵⁴ Erik ibid



some kind of residual vision and it can say how the light usually feels or something."

Johanna⁵⁵: *"Yes, actually mostly using the smell and feel of the hand itself and stuff like that. So it's regardless of whether you're blind or not, so it's good when you buy something to smell it and feel the texture of it...It is something good for the common person, but for us blind, it is the only thing that can be a thing to value the food if you could say so. Because yes, because the smell, it's mostly the smell. Then you can understand which departments you are in, then it is sometimes easier..."*

The majority of the informants displayed a personally developed understanding of their senses interplay in their retail experience. The statements above support both Yakhlef (2015) and Krishna and Morrin (2008) implications of perception through senses. For Yakhlef (2015) it is on the improbability that one singular sense can capture the whole experience and for Krishna and Morrin (2008) it was about the use of other sensory cues to build upon one's perception. One interesting remark is that regardless of residual vision, the sense was still being used in the retail experience. Helmefalk and Hultén's (2017) research on visual density and the complementary cues of auditory and olfactory as enhancing tools does matter more in this case. VIP do suffer from different kinds of sight conditions, but they might still use their vision somewhat to perceive. By having other sensory cues as complementary it can strengthen VIP sensory perception and improve their overall retail experience, it can greatly increase VIP perception than adding more visual cues only. One contrast that contradicts other literature and the empirical data comes from Spence et al. (2014), arguing that more than two congruent stimuli can actually negatively affect the consumer. But many of the informants describe an active use of several senses to create a mental image of the environment as well as navigating and recognising products. Therefore, adding more than two congruent stimuli in the retail experience would probably enhance rather than worsen for visually impaired customers, showing the unique perspective of VIP experience. These findings strengthen the importance of Altay's (2021) suggestions.

During the interviews, it became clear that the use of each sense varies depending on the specific situation which emphasises Helmefalk and Hultén's (2017) theory on the varying relevance of each sensory cue. The auditory senses were mentioned by the informants as one of the most varied ones. According to the following empirical data, the auditory sense concerns itself with consideration of other consumers, orientation recognition, and product recognition. In the literature, there is a lack of data regarding the use of auditory as a facilitator for navigation, indicating a need for future research on the subject.

Anders⁵⁶: *"When you have a visual impairment or a person who has a visual impairment, you often always use your hearing. You are much, much stronger in*

⁵⁵ Johanna (n 3).

⁵⁶ Anders (n 43).



other senses...People also help, but without awareness. Which like you hear the people far behind. ...People, sitting and talking about which nice flowers there are. That's how I know there are flowers in there somewhere...."

Sofia⁵⁷: *"I actually sometimes think that I can find the right place in the store with partly different freezers and refrigerators sound in slightly different ways and the checkout noise so that you can find the exit..."*

Johanna⁵⁸: *"Most of the time you hear when someone, someone, some cart or a carriage pulls like that and I understand that it is people who are in front of us like that or when you move goods then you know that now there is someone with a big cart perhaps sorting things, so I have to be careful there, in such a way that then I am hearing."*

From these empirical findings, it was found that sounds and VIP's auditory skills were a substantial part of their perception and complementary tool for the overall retail experience. As stated by one of the informants above, other consumers contributed as integral parts of the retail experience through their presence. This is an example of how congruence from a customer-to-customer interaction occurred without direct engagement (Nguyen et al. 2020) as well as the importance of auditory cues that the VIP relies on (Helmefalk & Hultén, 2017; Altay, 2021). Additionally, previously mentioned in *Personnel Aids*, it was found that VIP also use their auditory skills to find personnel as well as *"find out which checkout is free."* (Johanna). In these situations, when the auditory sense is dominant, waiting was mentioned repeatedly by the informants as a connected aspect. VIP must be sure that the opportunity in front of them is correct for them to act accordingly, where the auditory sense is useful:

Johanna⁵⁹: *"Waiting a lot and the hearing to listening if there is someone personal sorting goods on the shelves. But if there is such a person, I can come to the sound where they are sorting and ask the person who is sorting and say. "Can you help me with it?"*

Anders⁶⁰: *"...when you use your hearing you try to find staff, in some way....be like you hear and there is someone like that. "*

As described above, the auditory sense enables a tool for orientation and navigation, a tool which several of the informants argue as an important facilitator during the retail experience. In environments that are familiar to visually impaired customers, different auditory cues such as the placement of speakers with background music provides indications on where the specific location is.

⁵⁷ Sofia (n 27).

⁵⁸ Johanna (n 3).

⁵⁹ Johanna ibid

⁶⁰ Anders (n 43).



Identification of the environment and the atmosphere can also be done by using auditory cues:

Erik⁶¹: *“ But it is also about sonar in some sense. Passive ekologization is what you say..., so that you have as well as you hear how. How sounds of different kinds bounce off walls, ceilings and floors and so on. So you can hear a bit what kind of environment you are in.”*

Therefore, implementing auditory cues will provide a more accessible environment. When entering a store, sliding doors are common. These doors make a sound, informing VIP that the door is open, ensuring to enter the store without crashing into the door. Implementing similar stimuli during other parts of the experience would be preferable for VIP since *“It's also quite good that you have, that it, that it's not so noiseless”* (Magnus). Similar to this, the majority of informants emphasise all the elaborate sounds that contribute to the VIP retail experience, adding to what has previously been mentioned about orientation in *Tactile Implementation*. From a sensory perspective, auditory senses can be used as orientation in the visually dense retail environment, possibly elevating VIP retail experience further (Helmefalk & Hultén, 2017). However, having auditory as the most varied sense can be problematic since it can be easily overwhelmed, which constitutes a possible disadvantage for VIP:

Magnus⁶²: *“Because most of the time there are a lot of people in, uh, the shops and then it can cause it to disturb, so that it disturb when you have to sort of hear where, but for example. For example if you are going into the shop it can be a little difficult to hear exactly where the door is because there is so much background noise or if you have to go out it can also be very difficult to sort of. Locate exactly what the door is.”*

Johan⁶³: *“Yes, so if there was a lot of noise, it could affect whether I have to sit and listen to descriptions. Because you have to, you don't always wear headphones either. That you want to hear your surroundings too. So, for example, if there was a lot of noise, it could have an effect.”*

Even though these findings are not identical, they describe similarities to what Spence et al. (2014) mentions regarding how too much stimulation can affect perception. The informants describe the risk of overstimulation which can cause difficulties for them. For example, when using technological aids during grocery shopping, the customer experiences difficulties choosing to hear either the aid's verbal description or the surroundings. Both are incongruent with each other and are high in their exposure simultaneously. These findings argue against additional sounds as a beneficial implementation in an already auditory crowded environment, an argument which is supported by Spence et al. 's (2014) findings.

⁶¹ Erik (n 2).

⁶² Magnus (n 15).

⁶³ Johan (n 1).



The sensory cues can also be seen as expected-relevant and unexpectedly relevant because of their richness in information, which can drastically affect the mood of the customer because of incongruent factors and can affect the overall VIP relationship within the retail space (Nguyen et al. 2020; Heckler & Childers, 1992).

There is a complicated multifaceted aspect of the olfactory in supermarkets, reflected in what the supermarket offers, big or small, in its selection. From the empirical data it can be perceived that the olfactory cues are not substantial in a way to facilitate its own response. For example, informants used it to reinforce their orientation in the supermarket by the smell of baked goods. However, it was mentioned as less reliable. These findings complement the factors stated by Kostyra et al. (2017) and Mau, Schweizer and Oriet (2023), indicating that not only physical implementations but olfactory cues affect the orientation and should therefore be considered when designing the retail experience for VIP. Additionally, olfactory cues should be considered as a possibility (Sandell, 2019) when designing a facilitating and accessible retail experience.

Erik⁶⁴: *"In stores that have a bakery section, for example, it's quite easy to orientate yourself to where it is somewhere. Or, or like a fish counter or something, so that in this way you can. You can, it can facilitate orientation, and then I don't know if it's something that would work in reality, as well as consciously making use of it."*

Johan⁶⁵: *"Yes, it depends ... Sometimes you can be deceived. Maybe you can smell, Here maybe it smells like bread, but maybe it's rather on the shelf, Or like that, It can be on the second aisle, but because they are next to each other or go parallel so sometimes you can be, maybe also misled . So it depends a little on how they are built and with compare so."*

Unlike the auditory sense, using olfactory senses as a complementary can be challenging which may entail incongruence for VIP. The following empirical data can further demonstrate this:

Olof⁶⁶: *"... You don't use that much (smell) in grocery stores because like everything is kind of packaged in a grocery store, almost always anyway. No, it's probably mostly the feeling"*

Erik⁶⁷: *"For example, when they have the bakery itself in a completely different part of, like, behind the store than where they actually have the bread sales. Then, that's how it is for some, at some of the larger stores and it's clear that it can be a bit confusing."*

⁶⁴ Erik (n 2).

⁶⁵ Johan (n 1)

⁶⁶ Olof (n 10).

⁶⁷ Erik (n 2).



There are numerous factors that make olfactory difficult or effective to use as a sensory tool (Sandell, 2019). This is the reason why sensory incongruence is important in this case (Eklund & Helmfalk, 2022). To properly pair the exact cues to the product, the right receptors are used. This is to decrease misleading perception and to increase product recognition, a process which needs to be facilitated for customers with visual impairment (Balconi, Acconito & Angioletti, 2022; Kostyra et al. 2017; Yu, Tullio-Pow & Akhtar, 2015). Incongruence can be used in any sense, but specifically differentiation can promote positive engagement from the VIP. There is a difference between small or big stores, if they have spacing on their shelf to begin with or if packaging is a substantial factor. Nonetheless, one informant made a good point about the use of coffee smell in retail settings:

Johanna⁶⁸: *"I know that for marketing purposes they use that coffee smell in certain places, even if there is no coffee there just because it is purely psychological as I have understood on some programs and so, so they do it to attract more people. Because you like the smell of baked goods and coffee a lot and so on, but you don't like the smell of fish, for example. You don't like that and then you try to avoid it in some way by locating in the fridge, having a wall so you don't have to feel those smells. But for mine, that is, the thing that's pretty good for a sighted person. But for me there will be some limitations because the clearer I am about how I feel the smell, the clearer the picture will be for me for the actual placement of food and goods and so on"*

Helmfalk and Hultén (2017) advocate for non-visual cues like auditory cues and, in this case, olfactory cues as supplements can be substantial for long-term customer satisfaction. However, as mentioned in the statement above, it can be problematic to add any kind of olfactory cue without thinking about the consequences. The retail environment should try to adapt its cues, in this case, olfactory cues to try to include and be accessible for others (Altay, 2021; Sandell, 2019). Furthermore, VIP identify specific cues with the sense of touch to recognize products when retail shopping. The empirical data found that there were some substantial limitations that made it difficult to rely on touch cues:

Fredrik⁶⁹: *"The sense of touch yes, I can feel the products too. It can be a bit complicated sometimes... So, I haven't come up with any other senses even to find things, but it's a lot of sight and kind of feeling around... Yes, when you get to the shelf so the things are at a little low levels, I usually feel it to see which products it is and so. If, it's a soft package or if it's hard or something like that"*

Sara⁷⁰: *"I always use my fingers for everything....Is all packaging, either paper or plastic. Then there can be a different structure on the plastic. Sometimes it's rough."*

⁶⁸ Johanna (n 3).

⁶⁹ Fredrik (n 6).

⁷⁰ Sara (n 12).



Sometimes it's blank so that, similar to boxes, but there I use my residual vision to identify the product by recognizing what the box looks like."

Touch as a sensory cue was one varied sense of its own, for example used as complementary recognition cue. Similar to both auditory and olfactory cues, touch can be used to facilitate the product recognition process. Additionally, one informant describes touch as an orientation tool to recognize the location in store. However, similar to Hultén's (2012) findings, the informants shared the perspective that the sense of touch relied on previously gained information. Additionally, these findings expand Helmeffalk and Hultén's (2017) theory of non-visual cues, providing a new perspective on touch as a sense which preferably needs to collaborate with other senses. As a retailer, enabling sensory cues that can be perceived by the sense of touch would ensure VIP to, for example, choose the right product and therefore ensuring the opportunity for independence (Kostyra et al. 2017; Guo & Bigham, 2018). There is a shared assumption that VIP do recognize with touch which is reinforced in the following empirical data:

Gustav⁷¹: *"I feel, I feel the tomatoes and, but there is not much you can feel. Most of it is packaged."*

Magnus⁷²: *"...when it comes to feeling that you should get the chance to feel different goods, products and I kind of see where yes okay it's this product... then maybe I don't have to ask so much next time, or things like that. So the chance to feel"*

The sense of touch is utilised in instances where products are previously known to be identifiable by touch evaluation (Hultén, 2012). For example, in the fruit area touch is used, but for packaged products it is about recognizing that product's material/form again for future references. As mentioned in one of the statements above, this approach is to avoid future inquiries that might occur. This aligns with Kostyra et al.'s (2017) explanation that VIP tends to purchase their groceries from already-acquainted brands when it comes to their decision-making process, avoiding unknown factors that may occur. To avoid this, adding more touchable cues could enhance the processes and enable VIP to explore more brands without the feeling of uncertainty, which could be achieved by adding congruent tactile cues (Helmeffalk & Hultén, 2017). These findings provide even more depth to the implementation of tactile elements, discussed earlier as beneficial in-store design suggestions. Additionally, in the following empirical data an important aspect for VIP in product recognition through touch is brought up:

Fredrik⁷³: *"Like when you pick up a canned food. It's hard to kind of feel it. I think you should feel it in a way. It is difficult to recognize the letters on the packages. In other words, that's kind of why I have to stand for a long time at a shelf and pick*

⁷¹ Gustav (n 17).

⁷² Magnus (n 15).

⁷³ Fredrik (n 6).



up one product at a time. To take a closer look at it... So, it can be like when it says a can with something like pea soup, to pick up can by can until I find pea soup, and you have to pick up several products before I've found the right one."

For the majority of informants there is an already obvious, clear-cut use for touch in their retail experience. From a perspective of touch, there is an apparent untapped potential. The statement above brings up an aspect where VIP suffers from canned food design, providing an example of a product design that can be elevated by having different incongruences to display differences between canned food products (Eklund & Helme Falk, 2022). Altay (2021) and Ekelund and Helme Falk (2022), as well as Krishna and Morrin (2008) discussion on how different consumers might perceive cues is important. If there was a way to apply accessibility to ease the product recognition, then it can give the insight for positive customer relationship foundation for VIP. Additionally, the majority of the informants describe grocery shopping as energy and time consuming, regardless of whether the individual needs to complement their residual vision or they need to completely rely on other senses.

When asked about their overall perspective on the retail experience in supermarkets, the informants answers varied but a commonality in the use of energy in relation to the use of senses was apparent. VIP avoids specific experiences not only for social difficulties but also because of their limitations of perceiving. This indicates a combination of sensory overload and absence of sensory cues (Spence et al. 2014). According to both Heckler and Childers (1992) and Eklund and Helme Falk (2022), the retail experience would be enhanced due to implementing congruence. Contrary to these studies, the empirical findings of this study do not show the same result but rather indicates VIP's non-existent observation of congruency. However, the subconscious impact of (in)congruence among VIP cannot be overlooked and should be further researched in the future.

Fredrik⁷⁴: *"Quite a lot actually, so when I have to buy a few goods, it's not that much, but when I have to buy a lot of goods, I'm tired in the head when I get home. So after you have to buy goods that you haven't bought before, I know, then I have to use energy to play for those goods and stuff like that"*

The statement above exemplifies VIP energy consumption further, stating that it takes a lot of energy to shop in large quantities, while other informants avoid events because of the sheer amount of visual impact. The informants pick and choose how they avoid the negative effects that come with sensory overload (Spence et al. 2014). Nguyen et al. (2020) explained their insight on industries like retailers when it comes to either high or low involvement in consumption, it becomes important to consider because of its position to enhance retail brand image through effective engagement with the customer base. In the case of high involvement, congruent customer characteristics can positively influence customer

⁷⁴ Fredrik (n 6).



satisfaction and purchase behaviour. But from a VIP standpoint, their congruent customer characteristics are not considered at all, and it has the consequences of negatively affecting VIP from a customer perspective. For the VIP this can be seen as overly high involvement from their end and can be seen as problematic. It becomes even more relevant when considering that VIP do not necessarily buy products or services but the experience (Srinivasan & Srivastava, 2010). It is an apparent struggle that Olof and Johanna even expressed as VIP in the following empirical data.

Johanna⁷⁵: *"This is precisely why I use companion because if I were to be myself and use my senses. It's going to be a lot, I'll even get a ache even in the head when I'm focusing so much."*

From this perspective, VIP are challenged in a way that they prefer having companions to ease the retail experience. Similar to the findings in theme *Personnel Aids*, this perspective emphasises the importance of a well-functioning implementation of personnel, not only because of the mentioned factors but also for reducing the energy consumption amongst VIP in supermarkets. These findings complement factors for positive service engagement (Nguyen et al. 2020) and increase the knowledge of senses impact on VIP as a customer group (Ekelund & Helme Falk, 2022). A customer group that is equally important as any other.

4.6 Updated conceptual framework

A conceptual framework was put together after the literature review to visualise how existing theories approach the phenomena. There the three blocks of in-store design, aids and senses were a part of the retail experience as their own independent concepts that VIP encountered when visiting supermarkets. During the literature review it became clear that the views of Dodds and Palakshappa (2022) are as relevant as ever. Companies view accessibility as something for people with disabilities and divide them from other types of customers. This is noticeable in the conceptual framework presented earlier that is formed from already existing literature. The conceptual framework has since been updated (See Figure 2) as new perspectives have been discovered by our research, providing a visualisation of how the three concepts are not only a part of the experience but interacts constantly with each other, shaping VIPs' perception of the experience as it does for any other customer. The three identified themes have also been added to the framework as they are integral parts of multiple experiences.

5. Conclusion

In this final chapter, the conclusion of this thesis is presented. Firstly, the research questions will be answered separately, followed by the implications from the

⁷⁵ Johanna (n 3).



perspective of theoretic, managerial, social and ethical as well as sustainability. Lastly, the limitations and findings for future research will be presented.

5.1 Increasing accessibility in Supermarkets

Offering an accessible customer experience for all customers regardless of disability is more relevant than ever, especially when it comes to something so vital as shopping grocery products. VIP demands retailers in the supermarket industry and developers of facilitating aids more knowledge of how to facilitate as well as effort to implement enhancing accessibility factors. Existing aids for VIP are not adapted for the intended task (Plikynas et al. 2020; Boldu et al. 2020) and strategies such as temporary product placement for increasing customer attention (Drèze, Hoch, & Purk, 1994) constitute major barriers for VIP's retail experience in supermarkets (Kostyra et al. 2017). Sensory aspects are becoming increasingly relevant (Yakhlef, 2015; Helme Falk & Hultén, 2017; Nguyen et al. 2020), however, research on the impact of these aspects for VIP is non-existing. Therefore, this study has aimed to explore how VIP experience supermarkets and how in-store design, senses and aids impact the experience as well as how these factors affect each other. This has been possible for the study as three main themes that all relate to each other have been analysed. The themes which were identified during the thematic analysis work as a broad foundation for answering the two research questions. Consequently, the two research questions will be answered separately in order to give a comprehensive and nuanced answer.

5.1.1 How do supermarkets in-store design and sensory cues affect the visually impaired's retail experience?

The first research question is best answered by the two identified main themes of *The Physical Space of Supermarkets* and *Vip's Senses in Retail Experience*. *The Physical Space of Supermarkets* is divided into two sub-themes of *Layout, Placement and Visuals* and *Tactile Implementation* to better understand how in-store design affects the visually impaired's retail experience. Supermarkets in-store design can affect the retail experience for VIP in both positive and negative ways depending on how it is implemented. In-store design that is thoroughly thought through with VIP in mind enhances the experience and allows for a smoother visit. The factors that facilitate this includes avoiding rearranging and moving products, whereby having permanent placement of shelves, displays and products the experience becomes less challenging for VIP. The same impact is found in supermarkets that utilise clear and wide aisles. The in-store design could further affect the retail experience for the better for VIP by implementing tactile walkways in supermarkets, tactile markings on various products as well as tactile maps in connection to the service desk. These concepts would enhance the accessibility in supermarkets and heighten the independence for VIP.

In-store design is also shown to affect visually impaired consumers differently from person to person. Not every individual who is visually impaired experiences



in-store design the same in supermarkets. This is notable in the impact of lighting and signs, where VIP desire distinct contrasts for a more pleasant and accessible experience. However, as this research shows, at this point in time there is no universal contrast or colour combination that is able to elevate the experience in supermarkets for every single consumer that is visually impaired.

To answer the question of how senses cues affect the retail experience the theme of *Vip's Senses in Retail Experience* was identified as a significant part. Being a visual impaired customer requires different sets of needs and opportunities that supermarkets fall short on. Having visual impairment or even blindness supports the perspective that all other sensory cues matter significantly. In supermarkets, sensory cues have been aspects that matter for the VIP. From overall cases of sensory congruence and incongruence to specific cues like auditory, olfactory, touch, and even visual cues, VIP could substantially benefit from research on complementary perception through the various senses and how they interact with each other, congruent or incongruent.

From the overall sensory aspect, it was shown that VIP do use all of their senses such as auditory, olfactory and touch to perceive all around them, but they still rely on their visual perception. Even if more than two stimuli are being used by the VIP, they are required to be able to perceive accurately. In the auditory aspect sound cues were significant for VIP orientation in relation to their environment, other customers and personal. It was also perceived as a potential complementary cue to visual dense circumstances. Negative aspect that can make it difficult to rely entirely on auditory sense and cues is that it can be overwhelmed. This can also affect VIP usage of auditory assistive tools and overall mood because of incongruent cues. From the olfactory sense it was also shown that this sense and its cues were as multifaceted as auditory but was mentioned as accompanied sense and not entirely reliant. A susceptible sense which makes it difficult, but vulnerable to incongruency in for example, product perception. The sense of touch is a limited sense. It is seen as a complementary cue to recognition and somewhat orientation. The use of touch is usually used for reinforcing already familiar information. But touch in retail space is limited and has the possibility to improve beyond its present position. Lastly, by being visually impaired, they experience excess use of energy, sensory overload and absence of sensory cues, which can be prevented by offering well-functioning personnel aids in store.

5.1.2 How do aids contribute to the retail experience for visually impaired people in supermarkets?

To provide a nuanced answer to the second research question the theme *Crucial Aids* were identified, providing insights and a deeper understanding of aids for VIP. In terms of facilitating the retail experience for VIP through aids, two sub-themes need to be considered: *Physical Aids* and *Personnel Aids*. The most useful aid today is the telephone and its visual interpretation apps even though these are far from optimal for their purpose, the consequences of which are mainly irritation,



discomfort and time as well as energy consuming. Instead of enabling independence, existing aids do not fulfil the needs in a successful way, contributing with more difficulties than advantages. To improve the experience, a creation of a product database and additional in-store structures with coordinates such as aisle and shelf numbers should be considered. Stores could also utilise their own applications. Allowing customers to search for a specific product, providing coordinates as well as a navigation service to be able to easily locate a product. This service would not only benefit VIP but all customers. Additionally, using different audible facilitators for product recognition and self-checkouts would prevent uncertainty as well as reduce time consumption while grocery shopping.

Regardless of the physical aids, the need for personnel aids in supermarkets cannot be ignored. The availability for locating personnel and receiving help while grocery shopping mainly influences the choice of store. During the study, it was found that for VIP, the personnel play a crucial role during the visit in supermarkets. Contrary to earlier research, the findings indicate that VIP does not experience unpleasant treatment from in-store personnel. Three beneficial factors for facilitating the retail experience through personnel aids have been identified: attention from personnel; treatment and education for personnel; additional shopping service. However, to accomplish an accessible and beneficial retail experience for VIP, combining both physical and personnel aids is of great importance.

5.2 Theoretical implication

From a theoretical perspective, the findings of this thesis indicates that the retail experience for VIP is, in several aspects, an unexplored area of research. Analysing how the different aspects simultaneously affect the retail experience and combine in a successful way, the enhancement for accessibility is identified. For society, the importance of continually enhancing accessibility for VIP is crucial, as access to essentials like food from supermarkets is fundamental to daily living. As shown in the problem discussion, earlier research has researched in-store design, aids and senses as separate concepts of the retail experience and therefore providing implications without the aspect of how these impacts the other concepts.

When researching from a visually impaired customer perspective, this thesis has contributed with knowledge regarding a need for developing retailers' approach towards customers with disabilities and how this successfully could be done. Therefore, various solutions and implications facilitating the retail experience in supermarkets for VIP are presented, addressing the area's research gap. These solutions and implications show the need for seeing all factors as connected to each other as well as changing the perspective on in-store design from a visual-design-first perspective to a more nuanced combined-senses-design perspective. Researchers must abandon the old way of thinking of design as primarily visual, adopting the idea of design as a sensory tool involving several senses where the



creativity and management of the tool is the main point. Addressing the research gap concerning the needs and challenges faced by VIP in supermarkets will also enhance their independence. Finally, theoretical implications for increasing accessibility and independence for VIP in the form of in-store design implementations, suggestions for future development of aids as well as the different crucial aspects of senses have been presented, addressing the multifaceted challenges identified in the problem discussion.

5.3 Managerial implications

From a managerial perspective, this thesis provides insights into how the retail experience can be managed to facilitate accessibility and the importance of addressing the needs of customers with disabilities. As a retailer, understanding how subjects such as in-store design, aids, and sensory aspects can be managed to enhance the shopping experience for VIP is crucial for meeting the requirements of being accessible spaces. This customer segment constitutes a demographic that retailers must consider when designing shopping experiences. Implementing in-store design elements facilitating orientation and navigation as well as enabling the use of aids increases the independence, reduces the need for assistance and provides competitive marketing advantages. Furthermore, the most significant factor during the retail experience for VIP is personnel. Managers should consider this thesis three beneficial factors when managing the personnel to meet and help customers with disabilities in the best possible way. In addition, the aspect of senses needs to be considered when designing the retail environment. From the overall perspective, each sense should have a prominent role and be designed throughout the supermarket, from auditory cues to olfactory, all the way to touch representation. They can be used to reinforce congruent stimuli and differentiate via incongruent stimuli. This can facilitate and elevate the overall experience in the retail supermarket space.

The knowledge gathered from this research extends beyond the specific themes investigated in the report. It sheds light on the difficulties visually impaired individuals encounter in tasks that sighted individuals may take for granted, both within and outside retail spaces. Retailers need to address the aspects of in-store design, aids and senses as strongly connected factors which need to cooperate to achieve the desired result.

5.4 Social, Ethical and Sustainable Implications

Social, ethical and sustainable implications have been closely intertwined concepts when considering the position that VIP is placed in today's society, specifically in the supermarket retail sector. In terms of social implications, a large part of this research's focus comes from the notion of accessibility. To include those with disabilities a chance to partake in everyday life tasks, in this case, retailing in supermarkets, without feeling excluded entirely. Ethically, this research implies the



consideration of other individuals' experiences. Their perspective is also crucial, especially considering supermarkets' general societal position. The sustainable implications present the subject of social sustainability, the position of a long-term change of accessibility in the retail space. Retailing in supermarkets should be sensible enough that people with disabilities can feel like they have a place to conduct their daily needs without friction.

5.5 Limitations and future research

As there exists an extensive research gap in the field of accessibility and how VIP experience the retail setting, future research is necessary to further improve the accessibility beyond the retail setting of supermarkets. This study is merely a beginning to understanding how VIP experience daily tasks in the retail world. Additional aspects to the retail experience should be explored to get a deeper understanding of the phenomena from the perspective of VIP as retail experience consists of much more than what this study could cover.

Sadly, not all data could be presented in the study as both time and formal requirements did not allow for all statements to be cited. In Appendix G it shows many of the statements and codes found during the collection of data. A large part of the codes were potential themes and part of the retail experience for VIP but had to be de-prioritised since they did not develop or built upon the purpose of the research. Future research is therefore crucial to further improve and expand on accessibility for both VIP but also for other disabilities as well. Each theme and topic discussed and presented in the study could be studied further from other perspectives as there always exist more viewpoints. This study has researched the chosen topics from a conscious consumer perspective with the gathered data. This leaves a gap in the covered themes in how they could potentially affect the retail experience for VIP in a subconscious way.

Future research could also investigate the retail experience for VIP from a managerial perspective. Managers of supermarkets and other retail settings could potentially have other viewpoints and face different challenges than that of customers. The question of how to allocate the right resources or how to practically implement the findings of this study is something future research could delve deeper into. As society's view on inclusivity and accessibility is constantly changing, it becomes clear that this study is not the completed puzzle but rather a link between various pieces. Similar views can be applied to the presented updated conceptual framework this report visualises. Being updated is not the same thing as a final version or definitive answer to the subject and with further research down the line an even newer and more accurate representation will hopefully be created.

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Appendices

Appendix A - Table Socialstyelsen Classification

Tabell Socialstyrelsens klassifikation, kod H54 Synnedsättning inklusive blindhet (i ett öga eller båda ögonen) <https://klassifikationer.socialstyrelsen.se/> hämtad 10/4 2024, finns inget senast uppdaterade eller författare

Synnedsättning Kategori	Synskärpa med befintlig korrektion (egna glas)	
	Mindre än	Lika med eller bättre än
Lindrig eller ingen synnedsättning 0		6/18 3/10 (0.3) 20/70
Måttlig synnedsättning 1	6/18 3/10 (0.3) 20/70	6/60 1/10 (0.1) 20/200
Svår synnedsättning 2	6/60 1/10 (0,1) 20/200	3/60 1/20 (0,05) 20/400
Blindhet 3	3/60 1/20 (0,05) 20/400	1/60* 1/50 (0,02) 5/300 (20/1200)
Blindhet 4	1/60 * 1/50 (0,02) 5/300 (20/1200)	Ljusperception
Blindhet 5	Ingen ljusperception	
9	Obestämd eller ospecificerad	

* eller fingerräkning vid 1 m



Appendix B - Classification of Visual Impairment

Norrbottnens läns landsting. (n.d.) Klassificering av synskada.

<https://vis.nll.se/process/ward/Dokument/ward/V%C3%A5rdrutiner/Diagnos/ICD10/Klassificering%20av%20synskadekategorier.pdf> (Accessed 2024-04-10)

Rutindokument		
Arbetsområde, t ex klinik	Godkänd av	Sida 1 av 2
Godkännadatum	Ansvarig för uppdatering	Version

Klassificering av synskada

WHO:s synskadekategorier och omarbeting av Eva Lindstedts synnivåer

Synprestation	Synsfär	Synskärpa	Kategori	max mindre än - min synskärpa	Benämning av synhandikapp
Läser vanlig text på normalt läsavstånd. Ser TV och svarta tavlan.	Ej märkbart reducerad	> 0,3	0	< 0,8 - 0,3	lätt synnedsättning
Mycket små leksaker och föremål. Byggsatser Detaljer i bilder. Trä pärlor. Sy större stygn. Strössel, smulor. Bokstäver, siffror, ej liten text "Titta-nära"-beteende. Ser inte bilnummer på långt håll. Sätter sig nära TV, vissa problem med att se på svarta tavlan i skolan.	Rätt god men ej detaljer på långt håll	> 0,1 - 0,3	1	< 0,3 - 0,1	mättlig synsvaghet
Mindre leksaker, Kulor och bollar. Pussel och större bilder. Kortspel. Lego, Russin, flingor, chokladlinser. Strössel klaras ibland.	3 - 4 m	> 0,05 - 0,1	2	< 0,1 - 0,05	uttalad synsvaghet

NORRBOTTENS LÄNS LANDSTING

Rutindokument		
Arbetsområde, t ex klinik	Godkänd av	Sida 2 av 2
Godkännadatum	Ansvarig för uppdatering	Version

Ledsyn ganska bra. Bollspel kan vara tveksamt. Cykling går ibland. Vid synpuppfigter på nära håll: "Titta-nära"-beteende.					
Känner igen ansikten på nära håll. Stora leksaker, t.ex. mjuka djur, dockor, bollar, matservis, nappflaska, bestick, möbler, kläder. Ledsyn, tveksam, nivåkillnader svårt. Som regel ej bollspel.	0,5 - 1 m	0,02 - 0,05	3	< 0,05 (Fr 3 m) - 0,02 och/eller synfältsinskränkning 5-10° från centrum (10 - 20° Ø)	grav synskada med små synrester
Känner igen ansikten på mycket nära håll. Stora, färggranna föremål. Ledsyn tveksamt	< 0,5 m	P - 0,02	4	< 0,02 (Fr 1 m) - P (ljusperception) och/eller synfältsinskränkning med central rest < 5°, 10° Ø	grav synskada med mycket små synrester
	Blind		5	ej P	grav synskada inga synrester
			9	osäker	



Appendix C - Operationalization

Concepts	Interview Questions	Reasoning	Supporting Theory
The overall experience	3	This question gives the participant an open start to describe the experience and what they deem as important without us influencing with earlier asked questions as well as to get a general view of the visually impaired's experience in supermarkets. All perceptions are individual.	Dodds and Palakshappa (2022), Sandell (2019), Eklund and Helmfalk (2022)
Influencing factors	4-6	To get a wider understanding of which factors influence the experience for the visually impaired as well as how this influence takes place. To get a more nuanced picture of what affects positively and what affects negatively and why.	Kostyra et al. (2017), Nguyen et al. (2020), Balconi, Acconito, and Angioletti (2022)
The use of senses	7-8	Due to the impairment or absence of one sense, these questions were asked to understand how the other senses are used and affected by numerous factors.	Yakhlef (2015), Altay (2021), Eklund and Helmfalk (2022), Nguyen et al. (2020)
Aids for the visually impaired	9-10	As aids can be found in many forms and used in many ways, the aim here was to get the interviewee's perspective on this concept, which existing alternatives function well or badly as well as what one had wished for in the future.	Guo and Bigham (2018), Yu, Tullio-Pow and Akhtar (2015) Kostyra et al. (2017) Celik and Yakut, (2021)
Type of visual impairment	11-12	To enable an analysis between different codes or themes where the type of visual impairment stands as a factor as well as giving the participant a chance to explain how their visual impairment affects their everyday life.	Kostyra et al. (2017)
Other	1, 13-14	These questions provided supplementary information which the participant considered valuable as well as following the thesis's ethical considerations.	



Appendix D - Interview guide - English version

Introductory questions

1. Do you have any questions before we start the interview?
2. Do you agree to us recording this interview?

The overall experience in supermarkets

3. How do you experience your visit to supermarkets?

Influencing factors during the experience

4. Can you name the factors that you consider to be most important for visually impaired people when moving around in supermarkets?
 - *How do they affect you?*
 - *Are there any more?*
5. What factors affect the visit to supermarkets negatively for you as visually impaired?
 - *How do they affect you?*
 - *Are there any more?*
6. What factors affect the visit to supermarkets positively for you as visually impaired?
 - *How do they affect you?*
 - *Are there any more?*

Senses & identification during the experience

7. How do you identify products when shopping at supermarkets?
 - *Can you use any other senses or aids to make it easier to find products?*
8. Which senses do you use most during your visit?
 - *Is there anything that has a positive effect?*
 - *Is there anything that has a negative effect?*
 - *How do you feel that your senses are in harmony with each other?*

Aids during the experience

9. Which (external) aids facilitate the visit to supermarkets?
 - *Are there any specific aids that make it easier to move around the store?*
 - *Are there any specific aids that make it easier to find the right products in the store?*
 - *How does the staff affect your stay in the grocery store?*
 - *Are you missing something that you would like to see in the future?*
10. How would braille affect your visit to supermarkets?

Personal questions

11. Would you like to tell us what type of visual impairment you have?
 - *How does it affect you?*
12. Would you like to tell us how long you have had your visual impairment?

Closing questions

13. Do you have anything else you would like to add?
14. Are you interested in taking part in the study when it is finished?



Appendix E - Interview guide - Swedish version

Inledande frågor

1. Har du några frågor innan vi börjar intervjun?
2. Är det okej att vi spelar in denna intervju?

Den generella upplevelsen i matbutiker

3. Hur är din upplevelse av vistelsen i matbutiker?

Påverkande faktorer under upplevelsen

4. Kan du nämna de faktorer som du anser är viktigast för synskadade när man rör sig i mataffären?
 - *Hur påverka dom dig?*
 - *Finns det några fler?*
5. Vilka faktorer påverkar vistelsen i mataffären negativt för dig som synskadad?
 - *Hur påverkar dom dig?*
 - *Finns det några fler?*
6. Vilka faktorer påverkar vistelsen i mataffären positivt för dig som synskadad?
 - *Hur påverkar dom dig?*
 - *Finns det några fler?*

Sinnen & identifikation under upplevelsen

7. Hur identifierar du produkter när du handlar i mataffären?
 - *Kan man använda några andra sinnen eller hjälpmedel för att underlätta att hitta produkter?*
8. Vilka sinnen använder du mest under sitt besök?
 - *Finns det något som påverkar positivt?*
 - *Finns det något som påverkar negativt?*
 - *Hur upplever du att dina sinnen stämmer överens med varandra?*

Hjälpmedel under upplevelsen

9. Vilka (externa) hjälpmedel underlättar besöket i mataffären?
 - *Finns det några specifika hjälpmedel som underlättar att röra sig i butiken?*
 - *Finns det några specifika hjälpmedel som underlättar för att hitta rätt produkter i butiken?*
 - *Hur påverkar personalen din vistelse i mataffären?*
 - *Saknar du något som du skulle vilja se i framtiden?*
10. Hur skulle punktskrift påverka din vistelse i mataffären?

Personliga frågor

11. Vill du berätta vilken typ av synnedsättning du har?
 - *Hur påverkar den dig?*
12. Vill du berätta hur länge du har haft din synskada?

Avslutande frågor

13. Har du något mer som du vill tillägga?
14. Är du intresserad av att ta del av studien när den är klar?

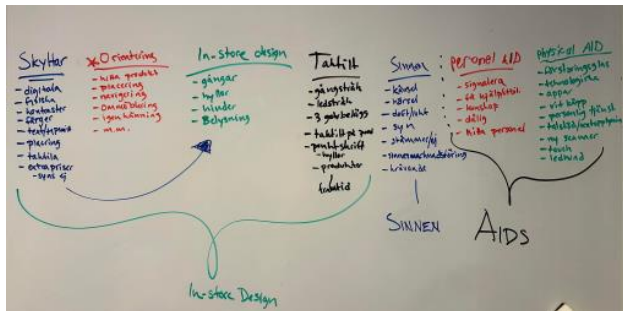


Appendix F - General information of sample

Anonymous name	Date conducted	Format of interview	Length of interview in minutes
Erik	April 15th	Teams	35:02
Anna	April 11th	Telephone	17:34
Johan	April 10th	Telephone	29:07
Johanna	April 15th	Telephone	38:47
Anders	April 11th	Zoom	38:05
Sofia	April 11th	Telephone	29:08
Fredrik	April 17th	Telephone	15:25
Magnus	April 15th	Zoom	20:24
Linnea	April 12th	Teams	28:45
Stefan	April 12th	Telephone	32:21
Sara	April 15th	Teams	30:28
Louise	April 16th	Telephone	25:54
Helena	April 17th	Telephone	15:03
Gustav	April 24th	Telephone	17:41
Olof	April 17th	Telephone	18:27



Appendix G - Thematic analysis phases



	Teknologiska	Det är nog svårt att få det tekniskt att funka utan personal tror jag
	Teknologiska	Jag har fått någon kamera ifrån syncentralen, men den kan jag inte hantera, för den är alldeles för bökig och trycka på knappar hit och dit så den klarar jag inte av... Och förstoringsglas, det hjälper inte
	Förstoringsglas	Och mina förstoringsglas, det hjälper inte och titta på priser. Det är för dåligt. Ja, jag ser för dåligt i förstoringsglasen för att kunna se priser helt enkelt. Det är mitt stora problem med butikerna.

IN-STORE DESIGN		
In-store design	Hinder	Alltså saker som är svåra till exempel, det är ju när man har hyllor som sticker ut i huvudhöjd när det inte är något som sticker ut i markhöjd, till exempel om du tänker, för att om man går med vit kapp eller man går med ledarhund, då upptäcker man ju i första hand hinder i i lägre nivå för att det är där man har kappen liksom. Man går inte att vifta med den runt huvudet så att, Att, det händer ju att man går in i hyllor som som sticker ut utan att det finns något ill marknivå om ni förstår vad jag menar.
	Hinder	men annonsdisplayer av olika slag som ställs ut liksom mitt i öppna ytor och mitt i gångar, och så som ibland kan vara lite rangiga om vi säger så om ni förstår vad jag menar. Det är lite jobbigt när man är liksom lyckas gå in i saker och välter ett berg av varor då för att de har staplat dem lite försiktigt
	Hinder	Jo, men det är ju återigen som sagt att inte ställa saker i vägen på dumma ställen så att säga som man kan riskera. För jag menar en en vit kapp jobbar ju, den ska ju hitta hinder. Du kommer ju liksom slå in saker med kappen. Det är ju det som är poängen med kappen att den slår in i dem för du ska veta vad de är och du ska inte gå in i dem själv, så då måste man ju se till att de saker som man ställer på golvet inte liksom riskerar att falla över och gå i kras alltför enkelt om man puttar till dem.
	Hyllor som hinder	Och likadant då det här att inte ha som sagt hyllor som sticker ut i huvudnivå då som kappen inte kan hitta till exempel.
	Orientering	Hyfsat liksom raka gångar att det inte är så mycket alltså att man tänker att man bygger lite på rutnåsar i någon utsträckning av det. Att man inte har för mycket, liksom gångar som går åt olika håll i olika riktningar, så utan att du kan lite lättare orientera dig.
	Belysning	belysning var sådana grejer som påverkar. Asså, om, det är en del av affären som är väldigt mörk eller om det är en del av affären som är väldigt ljus så är det saker som kan göra att det är svårare, att orientera sig. Att det är svårt att hitta hitta varor och så.
	Belysning	Det är väl ljusförhållanden då för min egen del. Det är för ljus, jag är väldigt ljuskänslig. Jag ser väldigt Alltså jag blir nästan dagblind om det är för mycket ljus, det beror ju på vad det är för typ av affär. Och sen är det ju också, Det kanske inte är ljusare eller mörkare, men själva upplevelsen är sån
	Smala korridorer	Ja extra negativt om det är smala korridorer skulle. Det är enklare att röra sig om man skulle haft generellt breda. För att man, då är man. Om det är smalt och sen kommer kanske någon annan kund förbi eller så. Och har man synnedsättning eller om man är till exempel blind eller så går man med kapp så kan man, vägar inte komma för nära varorna för att inte stöta till dem.
	Känna igen produkter	Ja i de bästa världarna om man hade kunnat haft någon form av. Båda, namnet i stor stil samt om man hade haft i punktskrift. Om man skulle säga generellt i de bästa världarna, om man hade fått önska sig frött... Men annars kanske, stor stil känns det, bör man kunna göra.
	Hinder	Jag själv ser ingenting framför mig, det är som en dimma, men lite grann på kanterna stött så här så att, när jag går så jag ser lite grann att det är hyllorna på en någon sida så här, men jag tycker det är jättedåligt om man placerar vagnen eller vagnen eller såna saker mitt på vägen och så och eller såna ja, du vet när de när personalen i affären skulle sortera saker och så, då kan de då kan det handa mitt på vägen, står det i sånt det. Då kommer jag inte upptäcka den och stöta på den direkt. Eller när hyllorna baserade på så sätt så att varor nästan sticker ut så att man kan lätt träffa dem med axel och, jag är lite rädd för dom.
	Smala korridorer	en bra placering, det finns ganska breda, breda vägar så att man inte råkar stöta på någon någonting på en måx, för att det finns vissa affärer där man placerar jätte jättemått, alltså typ så att man blir tvungen att gå inte normalt så här. Man måste nästan gå med axel framåt, förstår du vad jag menar? Sån, sån är är jättedåligt det det finns, det blir såna situationer om man har sånt, jätte små eller små avlång affär eller jätteliten affär i centrumet. Det brukar vara så när man har liten utrymme för, men man måste sälja massor av saker.
	Belysning	Ja, det är faktiskt bra fråga till exempel jag upptäckte att i stora coopaffärer, de har lite mörkare typ av upplys, alltså lamporna och så men Lidl de är relativt ljusa och ja alltså vissa, vissa affärer de föredrar att ha lite, lite mörkaktigt stil i sina affärer. Men då faktiskt lite svårare för mig som har skilnad mellan ljus och mörker. Du har rätt, du har rätt i. Ja, faktiskt ljuset, för mig som är ser lite grann på kanterna, kan spela roll.
	Trånga korridorer	Det negativa helt enkelt det är nog just att, det är väldigt olika från ställe till ställe, men det är att en del är väldigt trånga och tajla så att man riskerar att vifta med någonting när man går där.
	Belysning	Det kan ju vara att det är ett mörkt hörn av en hylla, liksom att belysningen är dålig. Man har svårt att se, det är lättare att se tycker jag typ vid fryn och kylskåp, för där är ju det belysning i kylorna. Men ibland när man ser vid hyllorna där så kan det vara lite mörk. Så jag tycker det borde vara bra belysning i hela butiken. Men i dom butiker jag har varit i hittills så har det varit väldigt bra belysning. Men jag. Jag vet att det finns butiker i det där inte är så bra belysning. Det tycker jag är viktigt
	Smala korridorer	Det kan vara väldigt väldigt smalt. Så kommer det folk med så här stora kundvagnar och sånt. Och ja, det kan bli lite trångt så det kan bli problem, men då försöker jag ju ha med vita kapp och det hjälper ju oftast så dom flyttar på sig.
	Belysning	Fråga om hur belysning påverkar. Alltså påverkar ju inte så mycket egentligen, för jag liksom, jag tar ju inte hjälp av det så mycket sen märker jag ju av om lamporna släcks eller om dem redan är släckta eller om dom är tända liksom. Men det påverkar mig inte så mycket, liksom orientering eller under själva vistelsen
	Hinder	Att det inte står massa skit i vägen på gångarna. Som ställer ut saker som står i vägen
	Belysning	Det måste vara bra ljus. Mycket ljus och sen för det, ska det helst vara ledd, om man har leddplatta så inte flimrande lysrör och sånt... Nu har man ju kommit så långt för belysningen sitter ju inne i friskdiskarna. Det är ju inget som blandar, det får inte blanda ljuset. Det får ju, det ska vara indirekt ljus, alltså så man inte blir, ljuset ska finnas men du ska inte se det. Man ska inte se amatören om man säger så.
	Belysning	Jag har ju svårt. Jag måste ha mycket ljus in i en lokal om när det är lite halvdunkelt, då blir det precis mörkt. Då ser jag knappt inte nåt... jag behöver väldigt ljus om jag ska titta på något med förstoringsglas



Appendix H - Authors Individual Contribution

The following table illustrates the author's contribution individually and are based on the grading criteria from Guidelines for the degree project in Business Administration.

Author:	Amanda Funkquist (%)	Rasmus Lennkvist (%)	Musse Hassan (%)
Introduction	38%	35%	27%
Literature review	40%	25%	35%
Methodology	25%	38%	37%
Empirical Findings & Analysis	40%	35%	25%
Conclusion	35%	35%	30%
Academic language and communication skills	38%	37%	25%
Active participation in seminars	33.3%	33.3%	33.3%

List of Figures

Figure 1 - Conceptual framework

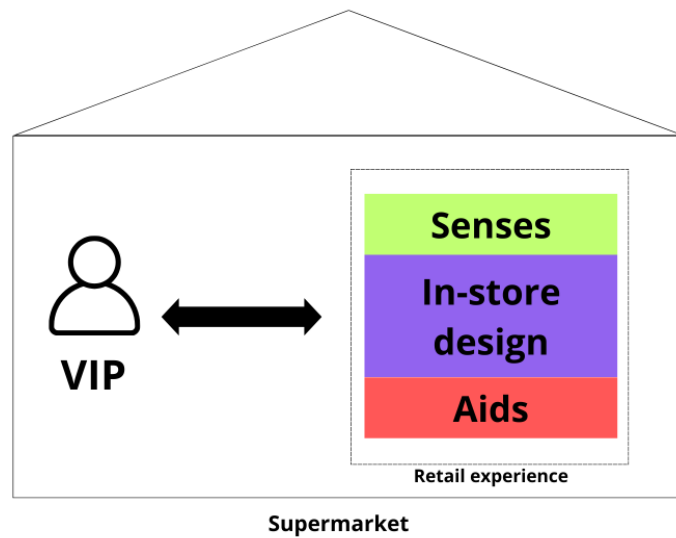


Figure 1: A figure of the conceptual framework of the retail experience for VIP in supermarkets (authors design)

Figure 2 - Updated conceptual framework

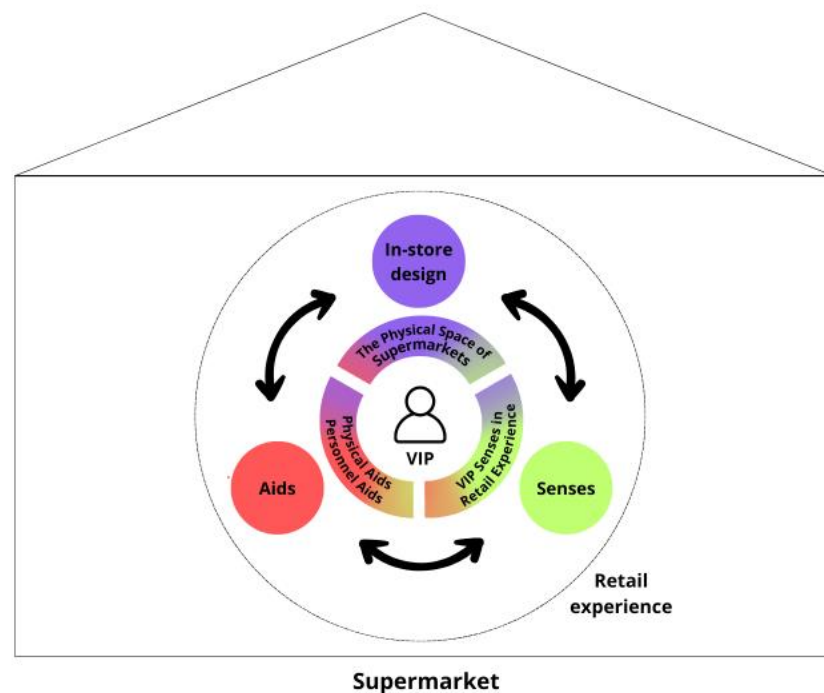


Figure 2: A updated figure of the conceptual framework (authors design)